



Development of an In-Situ Portable Hydrogel Bioprinter for Cutaneous Wound Healing

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Ctrl-Alt-Delete

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DEVELOPMENT OF AN IN-SITU PORTABLE HYDROGEL BIOPRINTER FOR CUTANEOUS WOUND HEALING

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ABSTRACT

Bioprinting is an emerging field in regenerative medicine that enables the layer-by-layer deposition of biological materials to create complex tissue structures. Despite its potential, current bioprinting technologies are often limited by high costs, poor portability, and bulky equipment that restricts their use to specialised laboratory settings. These constraints limit clinical application, where immediate and flexible tissue repair may be required.

To address this, the project explored the development of a portable handheld bioprinter for controlled bio-ink deposition at the point of care. The prototype incorporated a modular design powered by internal rechargeable lithium-ion batteries and controlled by an ATmega328P-based PCB, enabling self-contained operation. The device was designed with an octagonal cross-section to improve grip stability and reduce rotational slippage during handling, to support intuitive pen-like handling. Electrical validation demonstrated successful standalone motor control, with user-controlled extrusion and retraction implemented through a two-button interface. Mechanical testing confirmed stable actuation under low-load conditions. Initial system integration and assembled-base testing supported the feasibility of combining the electrical and mechanical subsystems within a portable handheld platform. Further work will focus on optimising extrusion performance and evaluating usability and clinical applicability.

1 Introduction

1.1 3D Bioprinting

Three-dimensional (3D) bioprinting is an emerging technology within regenerative medicine which enables the controlled deposition of living cells, biomaterials and bioactive molecules to fabricate tissue-like structures [1]. Developed from conventional 3D printing, bioprinting replaces standard materials with bioinks to produce biologically functional constructs with precise spatial organisation. Unlike conventional scaffold fabrication, bioprinting offers a high degree of customisation, enabling constructs to be tailored to the geometry and biological requirements of specific defects or tissue sites [2].

Its potential to address complex clinical challenges such as tissue damage, chronic wounds, and organ failure means that the technology has recently gained significant attention in bioengineering developments [3]. Through enabling the fabrication of customised scaffolds that mimic the structure and function of natural tissues, bioprinting can support cell proliferation, enhance tissue regeneration, and improve healing outcomes [4].

Specifically, *in situ* bioprinting, where biomaterials are deposited directly onto injured sites, can improve integration with surrounding tissue, and

reduce reliance on prefabricated implants or dressings [5]. Its ability to adapt to patient-specific wound geometries makes it a promising tool for personalised, regenerative medicine, where successful treatments depend on closely replicating the structural and functional characteristics of native tissue.

1.2 Current Limitations

The translation of bioprinting into routine medical practice remains limited. Most existing systems are large, technically complex, and confined to laboratory environments, restricting point-of-care use [6, 7].

These approaches are also limited by the mismatch between lab-printed constructs and patient-specific geometries [1], making them poorly suited to wound treatment where irregular geometry reduces scaffold fit and coverage [8]. Solutions to this usually involve extensive and expensive imaging modalities.

Handheld bioprinting has emerged as a branch of *in situ* printing that bypasses imaging needs, and addresses shape mismatch. However, existing systems still face important trade-offs, including:

- Ergonomic suitability for prolonged clinical use
- Precision relative to stationary bioprinting systems

- Standardisation and validation beyond prototype stage
- Portability for an otherwise satisfying system [5,6,9]

Challenges including flow-rate control, extrusion consistency, sterility, and electromechanical integration continue to limit device performance [10], while clinical translation is further constrained by regulatory requirements. This includes safety, reproducibility, ease of use, and compatibility with clinical practice [5]. No handheld bioprinter has yet achieved widespread clinical validation for routine wound-care, highlighting the gap between laboratory innovation and clinical need. Current research is therefore increasingly focused on developing portable, user-friendly, point-of-care systems.

1.3 Project Overview

The main objective of this project was to develop and evaluate a proof-of-concept handheld bioprinter for controlled bioink deposition in point-of-care applications. The device was designed to integrate electrical, mechanical, and usability considerations in support of portability and clinical ease of use.

To achieve this objective, the project involved:

- **Electronics:** The development of a motor-driven extrusion system.
- **Mechanics:** The design and fabrication of a compact 3D printed mechanical framework.
- The implementation of testing protocols to evaluate device performance and usability.

The aim is to translate bioprinting technology into an accessible tool supporting future regenerative paradigms, particularly in the treatment of complex and non-healing wounds.

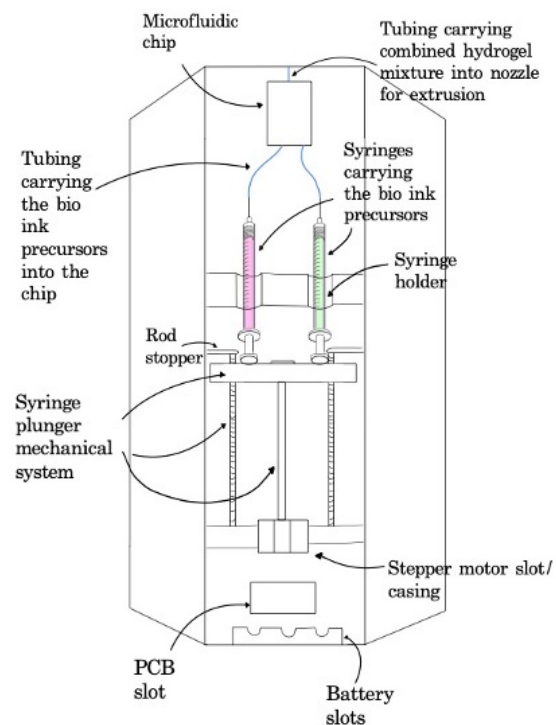


Figure 1: Schematic diagram of the handheld bioprinter, showing the integration of the microfluidic chip, dual-syringe system, tubing network, and stepper motor-driven plunger mechanism for controlled hydrogel extrusion, together with the PCB and battery housed within the device.

2 Requirements

The specification of the device needed to support performance of a gelatin-alginate hydrogel previously characterised by the lab. A microfluidic system needed to be incorporated with two-channel inputs and output tubing at 1m in length. The flow rate should be maintained at $1225 \mu\text{L h}^{-1}$ through a final 23G blunt needle for hydrogel performance [11]. Two syringes were needed for loading gelatin and alginate separately.

To push the syringe plungers, we opted for the standard stepper-motor lead screw system for its fine control over flow rate, compatibility with viscous biomaterial and accessibility. These combined gave us a general design to begin with (Figure 1), with detailed aims for design in the following sections.

2.1 Clinical Consultation

To inform development, a consultant clinician was interviewed to provide industry-informed perspective in addition to published literature. The consultation was used to assess whether a handheld bioprinter could address genuine unmet needs in wound care and to identify design requirements for clinical use.

Key topics included current limitations in wound management, the potential value of *in situ* biomaterial deposition, and the requirements for a clinically viable handheld device. These included portability, ergonomic handling and sterility considerations. The consultation also highlighted wider barriers in translation, including regulatory approval and need for further validation. The full consultation questions are provided in Appendix J.

2.2 Electrical Aim

- The system should drive the stepper motor to achieve target flow rate.
- The power management circuitry should regulate the battery output to a stable voltage suitable for the microcontroller unit (MCU), motor driver, and associated control circuitry.
- The PCB architecture should be compact enough to fit within the dimensional constraints of the handheld housing.
- The user input system should incorporate dedicated extrusion and retraction buttons.
- The electrical system should incorporate a main power switch.
- The programming interface should allow firmware upload and debugging during prototyping via an in-system programming (ISP) connection.
- The electrical layout should support accessible replacement of key components, particularly the battery and external switches for smooth maintenance.
- The bio printer must have hooks for the tubes on the walls to avoid obstructing components.

2.3 Mechanical Aim

- A rough upper limit was set on the dimensions: 300mm (length) x 55mm (height) x

50mm (width).

- The bioprinter should facilitate a neutral grip and optimal wrist positioning during operation.
- The barrel thickness should be within the ergonomically recommended range to prevent musculoskeletal strain (13-15mm) [12].
- Adequate weight distribution to prevent wrist strain.
- Ergonomic button placement to allow fingertip access during operation grip.
- Modular design for bio-ink syringe refilling and battery replacement.
- Flat resting surface for stable placement on platforms.
- The final prototype must be printed using ABS, for its mechanical strength and high electrical insulation [13], providing structural safety for internal components.
- The bioprinter should contain a pen-like nib for controlled bio ink extrusion.

2.4 Testing Aim

- The design should generate positive feedback from the given questionnaire to volunteers.
- The motor-actuation system should generate sufficient force to achieve extrusion under low resistance and representative loading conditions.
- User input should be translated into extrusion with minimal start/ stop delay.
- The integrated system within the base should demonstrate reliable coordination between all subsystems.

3 Methods

3.1 Electrical Design Methods

The main design considerations for the electrical subsystem were stable power delivery, reliable motor control, compact integration, and straightforward user operation.

The circuit was therefore divided into 5 main functional blocks that are separated modularly:

- Power management
- MCU Control

- Motor driving
- User input
- Programming

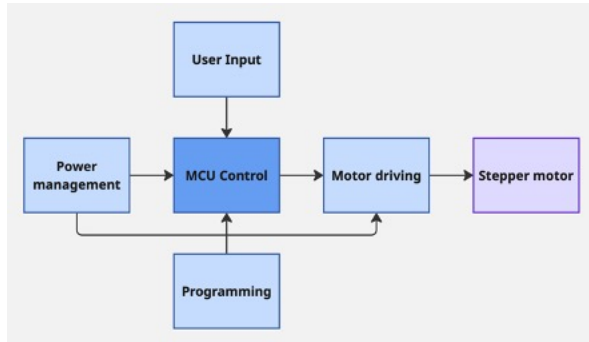


Figure 2: Functional block diagram of the electrical subsystem.

The power management block regulates the battery supply, while the MCU control block generates the signals required for motor actuation. The motor driver block links the control block to the stepper motor, and the user input block allows users to extrude and retract via buttons. The programming block provides an interface for firmware upload and iterative development during prototyping (Figure 2).

Component selection was guided by the trade-off between the dimensional constraints of the handheld casing and the electrical requirements of the actuation system. A compact microcontroller, stepper motor driver, and voltage regulator were therefore selected, together with an external crystal oscillator and filtering capacitors [14,15].

The circuit was initially built on a breadboard for isolated component testing. Following this, the circuit was then designed as a two-layer PCB, with an ISP interface for code upload [16].

3.2 Mechanical Design Methods

The mechanical design was split into 3 main sections:

- The outer casing forming the pen body.
- An inner base upon which the internal components are slotted.
- A backing plate containing the battery, and a nozzle for the tubing with hydrogel to go

through.

All the iterations were carried out in SolidWorks. The first design for the base was printed in PLA, for component fit, while subsequent iterations, including the final prototype, were printed in ABS.

The extrusion and retraction mechanism is controlled by the stepper motor whose casing is connected via a threaded rod to a slab (plunger) which are all placed on the internal trapezoidal base. Motor rotation would linearly advance the slab via the threaded rod, actuating the syringe plunger. Parallel smooth rods pass through the slab on either end, guiding its motion and preventing rotation. Pressing the corresponding button would trigger the motor to drive the slab forward or backward as required.

3.3 Testing Methods

A series of tests were conducted to assess the performance of the handheld bioprinter. Testing focused on:

- Electrical performance: voltage stability, current response, waveform quality and motor-control behaviour.
- Mechanical actuation and extrusion: low-load extrusion using water-filled syringes and integrated testing under hydrogel loading.
- Start/stop latency testing between button input and plunger motion.
- Questionnaire-based user evaluation.

A structured questionnaire was used to assess perceived usability, ergonomics, and clinical relevance following a brief introduction to 3D bioprinting and a video demonstration of the prototype and its intended operation. Responses were collected using five-point Likert-scale questions addressing:

- Ease of use
- Ergonomics
- Interface intuitiveness
- Point-of-care suitability
- Adaptability to irregular wound geometries
- Perceived clinical potential

A record of the questionnaire items is provided in Appendix J.

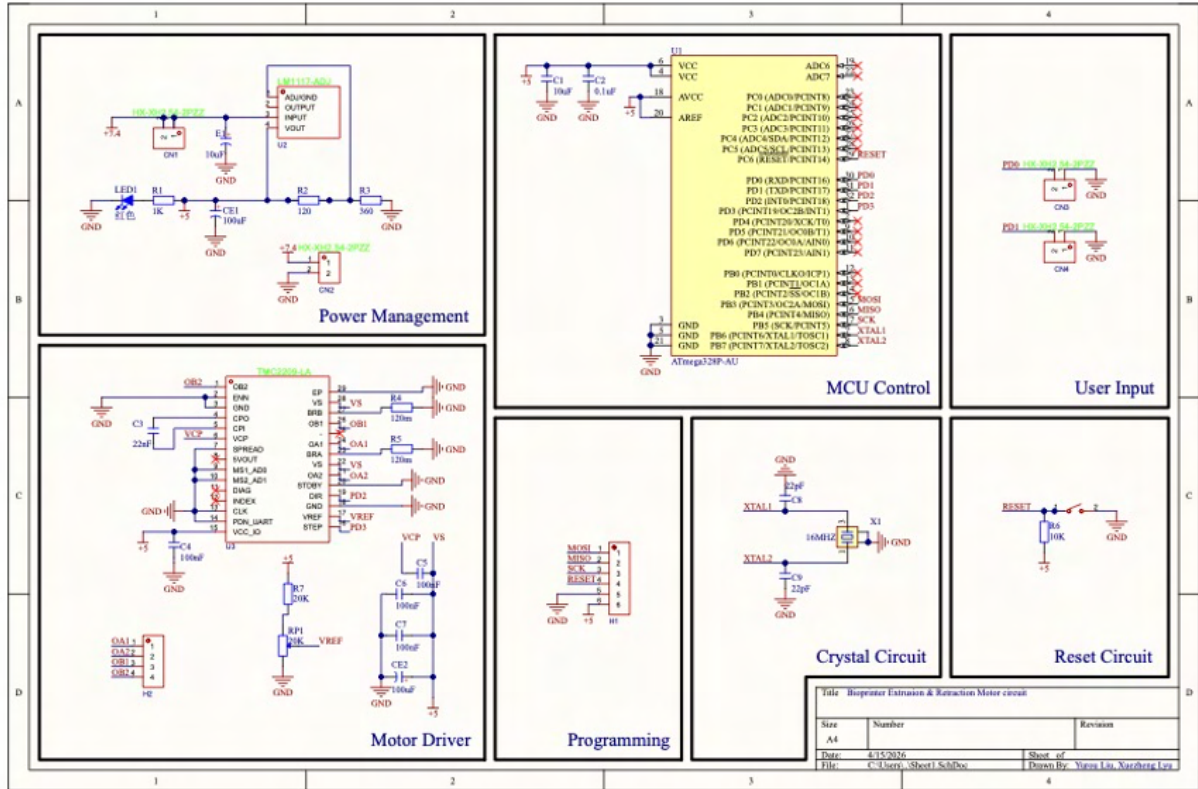


Figure 3: The schematic of the electrical circuit design, developed in Altium Designer.

4 Results

4.1 Electrical System Results

Final circuit architecture

The final electrical architecture (Figure 3) comprises of five main functional blocks: power management, MCU control, user input controls, motor driver, and programming. Supporting subcircuits such as the crystal circuit and the reset circuit were incorporated within the MCU control architecture. The system was powered by two rechargeable 10440 lithium-ion batteries (3.7 V, 750 mWh each) connected in series to provide a 7.4 V supply [30]. This was regulated to 5 V by an LM1117-ADJ voltage regulator for the logic and control electronics, matching the operating voltage required by the ATmega328P-AU microcontroller and the TMC2209-LA driver logic [14, 17, 18].

An ATmega328P-AU microcontroller supported by a 16 MHz crystal oscillator served as the cen-

tral control unit, while motor actuation was implemented through a TMC2209-LA motor driver. Electrolytic capacitors (100 μ F and 10 μ F) were incorporated into the power block to support stable circuit operation.

User control was provided through dedicated extrusion and retraction inputs, and an ISP programming interface was included for firmware upload during prototyping.

Breadboard-based functional validation

Staged breadboard validation was conducted prior to PCB implementation. The first stage, shown in Figure 4a, focused on standalone programming of the ATmega328P-AU using an Arduino Uno configured as an external programmer [29]. Successful firmware upload, together with the Blink test and stable power indication, confirmed correct operation of the standalone MCU configuration, crystal oscillator circuit, regulated power supply, and ISP programming pathway.

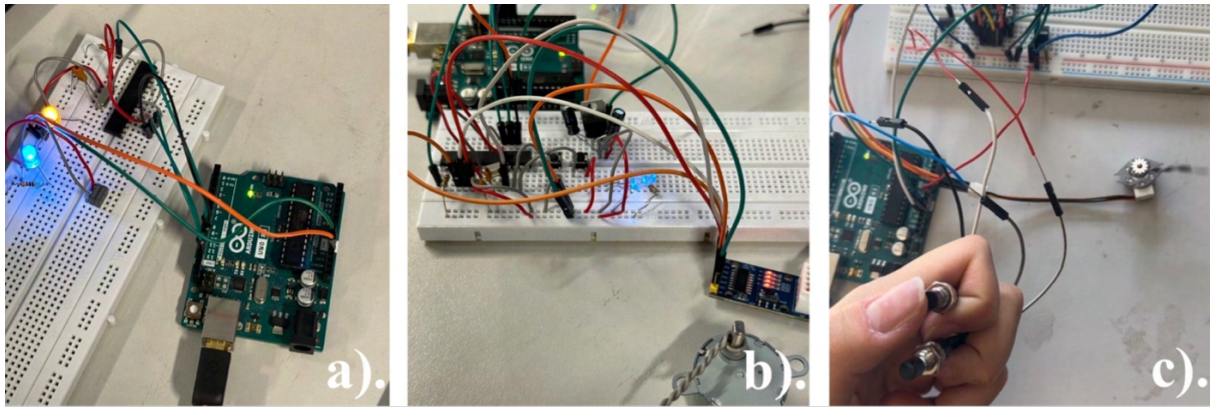


Figure 4: The staged breadboard testing for the electrical circuit design. (a). Breadboard setup for standalone ATmega328P-AU programming / validation. The blue LED indicates successful circuit power, and the yellow LED confirms firmware upload via a Blink test. (b). Breadboard setup for the core electrical architecture for motor function testing. The red LED on the motor driver indicates that the motor circuit was powered successfully during operation. (c). Breadboard setup for validation of extrusion and retraction control.

In Figure 4b, the core electrical architecture (MCU control, motor driver, motor) was added on the breadboard to assess motor actuation. Successful motor rotation using the initial control code confirmed that the microcontroller-to-driver interface and actuation pathway were operational.

In the final breadboard stage (Figure 4c), extrusion and retraction buttons were added to the breadboard, and the control code was further edited to implement forward rotation, reverse rotation, and stop functions. Together, these results confirmed that the overall circuit operated as intended.

Power delivery across the board was successful, indicated by the onboard green LED. Code upload to the ATmega328P-AU was successful as well; however, motor actuation was not achieved on this board version.

Further investigation traced the issue to accidental floating pins in the TMC2209-LA motor driver circuit. This led to abnormal heating during operation, ultimately damaging the motor driver.

First PCB iteration

The first PCB iteration was designed in Altium Designer (Appendix C), manufactured externally, and assembled.

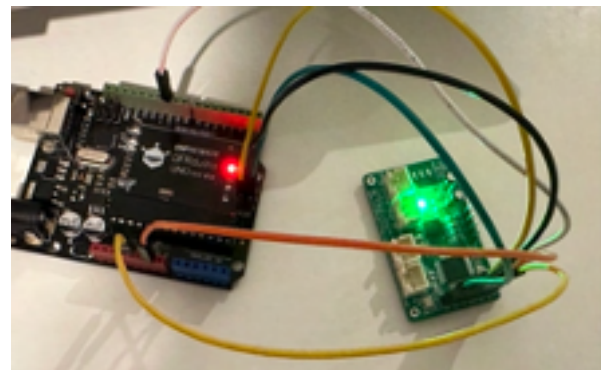


Figure 5: First PCB iteration connected to an Arduino Uno for firmware upload via the ISP interface. The green LED indicates successful power-on of the board during programming and initial testing.

Second PCB iteration

A second PCB iteration addressed issues identified in the first (Appendix C), containing revised TMC2209-LA schematic connections, a potentiometer for stepper motor current adjustment [19], and a heat sink on the motor driver to mitigate overheating.

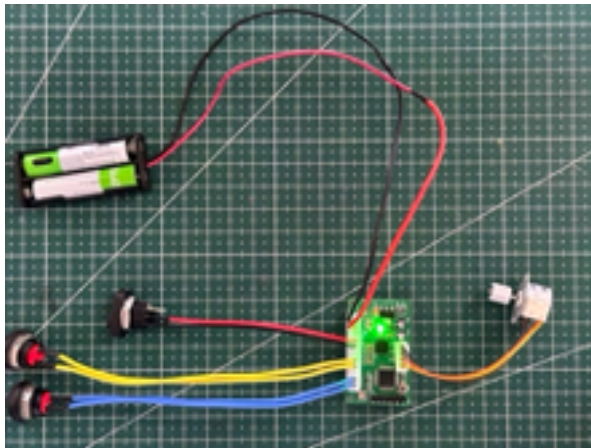


Figure 6: Second PCB iteration connected to the motor, battery, and external switches during integrated functional testing. The green LED indicates successful power on of the board during operation, and normal motor operation was achieved in this configuration.

Following fabrication and assembly, the revised PCB powered on correctly, indicated by the onboard green LED (Figure 5). Code upload was successful, and button-controlled testing confirmed normal motor operation, including forward rotation, reverse rotation, and stopping. Pressing the reset button during operation returned the micro-controller to its initial state and stopped the motor. The functionality of this PCB iteration was therefore fully validated.

Power consumption and theoretical runtime estimation

The electrical performance of the PCB system was tested under powered standby and motor-actuation conditions using a digital multimeter (Figure 7).

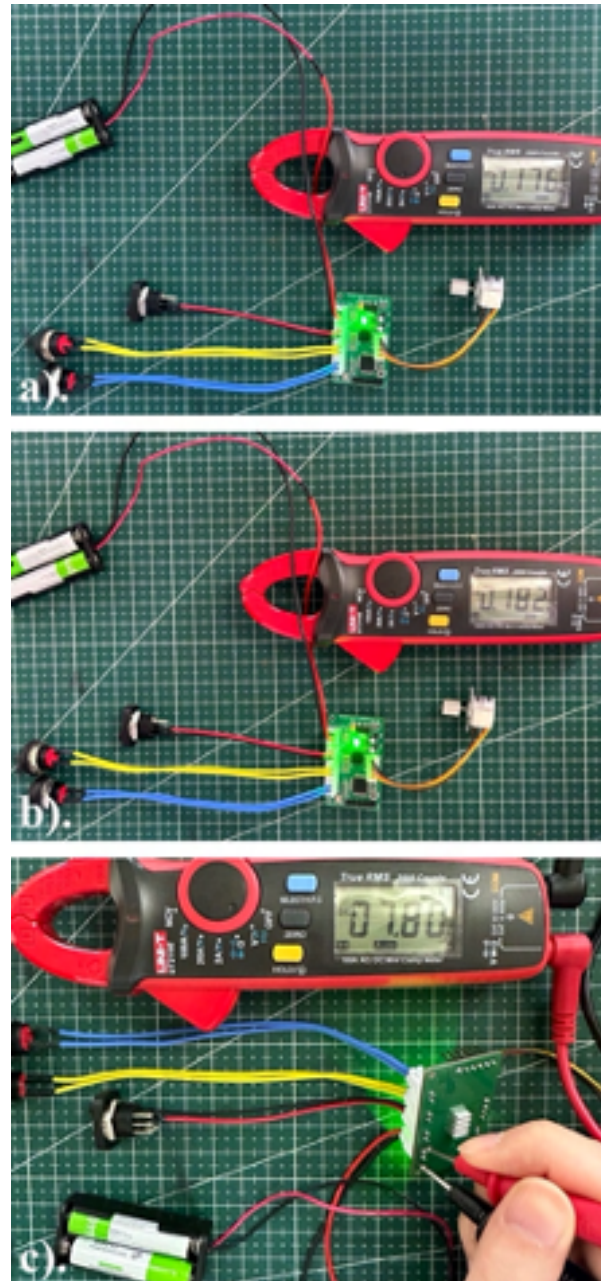


Figure 7: Electrical measurements of the PCB under different operating conditions. (a). Current measurement in powered standby condition; (b). Current measurement during motor operation; (c). Voltage measurement of the circuit during testing.

Operating condition	Voltage (V)	Current (A)	Power (W)
Powered standby	7.80	0.176	1.37
Motor operation	7.80	0.182	1.42

Table 1: Measured electrical performance under different operating conditions.

The current consumption remained close to 0.18A under both operating conditions (Table 1), corresponding to a power consumption of 1.37-1.42 W.

A slight increase in current was observed during motor operation, reflecting the additional load of the motor. These measurements were obtained using the earlier motor configuration rather than the final motor used in the completed prototype and should therefore be interpreted as an approximate indication of system power demand. Based on the measured power consumption under motor operation, the theoretical runtime of the device was estimated to be approximately 1.1 hours under continuous motor operation (Appendix D). However, this value remains a theoretical estimate only, as no full-discharge or long-duration endurance testing was carried out.

Arduino Code for PCB Control

Custom code for the ATmega328P-AU was developed [20]. A non-blocking two-button protocol using microsecond timers allows responsive extrusion control. To reduce accidental activation, motor actuation is initiated by a long press ($>1s$), whilst a short press stops rotation.

To achieve the target flow rate of $1225\mu L/h$, the code converted the required extrusion volume into motor pulses using a kinematic model based on:

- Syringe geometry, based on a cross-sectional area derived from a $1000\mu L$ syringe volume and 60mm stroke
- Mechanical drive characteristics, represented by a lead screw pitch of 0.5mm, matching a standard M3 rod to the motor's 3mm shaft [21]
- Motor resolution, defined by the 15PM20L02-SG100 motor's 200 steps per revolution combined with the 1/32 microstepping via the TMC2209 driver, ensuring smooth extrusion while remaining within the interrupt capacity of the 8-bit MCU [22].

This model yielded a pulse half-period of $383\mu s$, which was used to drive the motor at the required frequency of 1306Hz (Appendix E).

4.2 Mechanical results

Outer casing

The initial outer casing design was inspired by the rounded form and flat facet of the Apple Pencil

(Figure 8). The flat face was intended to provide a stable contact surface for mounting internal components, whilst a hinged opening on the curved surface allowed easy access and replacement of components. However, this concept was rejected because the hinge interferes with the flat base and reduce stability when placed on a bench (Figure 9).



Figure 8: Apple Pencil [23]

The next iteration replaced the rounded cross-section with an octagonal structure (Figure 10). This resolved the stability issue by allowing the hinged opening to be positioned on the upper face of the casing, leaving the base undisturbed. Although the shell width exceeded the ergonomic range outlined in Section 2.1, this was a necessary trade-off between compactness and the internal space required to house the device components.

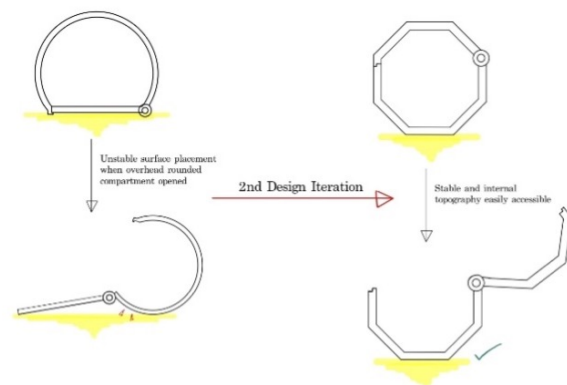


Figure 9: Iteration 1: Apple Pencil inspired casing. Showcases the issues with stability when the overhead hinged opening is fully opened. Iteration2: Octagonal outer casing with the overhead opening spanning the top half of cross-section resolving the stability problem.

Inner Base Structure

The inner base structure was designed to fit the lower half of the octagonal outer casing and provide a mounting platform for the device's main internal components, namely:

- 2 syringes carrying the bio-ink precursors
- A microfluidic chip for bio-ink mixing
- Tubing connecting:
 - i) Syringes to the microfluidic chip
 - ii) Microfluidic chip to the exit nozzle
- The stepper motor
- The plunger mechanism
- Batteries
- PCB
- Wiring

The structure is a hollow trapezoidal longitudinal support, with the cavity intended to house the batteries and wiring. Although the batteries were ultimately too large to fit within this space, the hollow design was retained to maintain lower weight, improving ergonomic handling.

Device Backing, Cover and Nib

A key design requirement was access to internal components for maintenance and replacement. The overhead opening allowed convenient access to components such as syringes and accessible tubing, and a removable backing was introduced to also provide electronic access by allowing the base assembly to slide out. The batteries are also mounted to the inner surface of the backing.

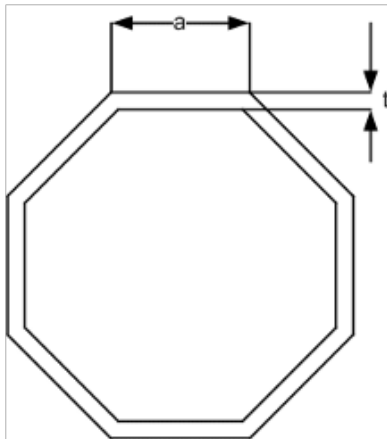


Figure 10: Octagonal Cross Section [24]

Partial CAD Iteration 1.0

The 1.0 iteration is composed of the main body of the bio-printer, divided into three parts:

- A top outer shell with a hinge to allow access to the syringe cartridge.
- A bottom outer shell forming the lower half of the housing.

- An inner base that allows motor, battery and PCB slotting.

1.0 Iteration Measurement	Dimensions(mm)
Outer shell length	271
Internal-base plan size	290 (length) × 30 (width)
Internal-base cross-section (rear end)	39 (width) × 30.04 (height)
Syringe pusher block	39 × 34 × 10
Motor-shaft diameter	Ø2.70 (×1)
Guide-rod diameter	Ø3.60 (×2)

Table 2: Key dimensions of Iteration 1.0 components.

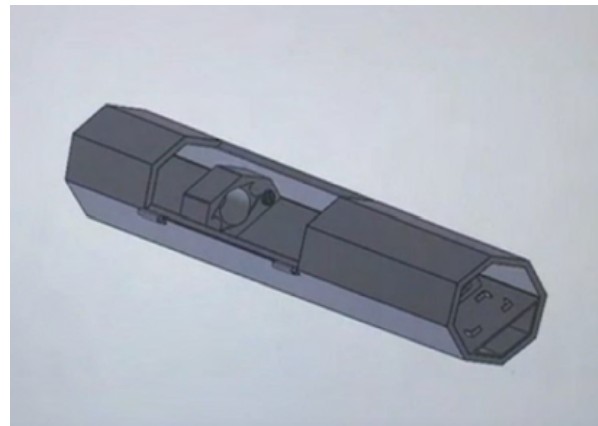


Figure 11: The CAD Assembly in Solidworks of the first iteration of the base and casing.

The 1.0 iteration required revisions since new batteries were chosen during electronics iterative testing, which no longer fit within the cross-sectional area. Furthermore, after 3D printing to test component fittings, the fitting tolerances were insufficient for the motor, microfluidic chip and PCB slots.

CAD Iterations 2.0

The inner cross-section was therefore increased from 46x46 mm to 55x46 mm through increasing the height. Additional functional features were also created:

- A tapered extrusion tip (pen nib) at the front end to guide hydrogel output.
- A removable backing panel for battery replacement.
- A solid button model on the outer shell to mark the motor control switch location.

Nib Measurement	Dimension (mm)
Base width (octagonal)	50.45
Tip bore diameter	Ø18
Taper length	59
Inner nesting depth	30

Table 3: Key dimensions of the extrusion tip (pen nib).

The button was modelled as a raised solid shape to indicate the intended switch position and size. These features were integrated into the existing three-part assembly structure (Figure 12). Sizing for base slots were also slightly increased.



Figure 12: CAD model of Iteration 3.0, annotated with the three functional additions: extrusion tip, removable battery backing, and solid button model.

After 3D printing and component matching, it was found that the tapered nib connected to the main body hindered tubing replacement and microfluidic chip slotting. The new height of the bio-printer also made it difficult to operate. A final revision is therefore required.

Final CAD Iteration

A final revision addressed assembly and component-fit issues previously mentioned with the following changes (Figure 13):

- Making the nib a separate part.
- The rear end only was raised to create a higher back profile while keeping the rest of the body at the initial height.
- Replacing the solid button model with cutouts to fit the physical switch.

- Enlarging the hinge opening and the corresponding flap to improve syringe access.
- Increasing the hinge dimensions for smoother rotation.
- Resizing the battery backing panel to correspond to rear height change.

Backing Measurement	Dimensions (mm)
Panel outer width	62
Panel depth	10
Internal octagonal cutout	42 (width) × 49.97 (height)
Wall thickness	2
Corner chamfers	8 at 45°

Table 4: Key dimensions of the battery backing panel (final iteration).

Technical drawings for each part of the final iteration are provided in Appendix H.

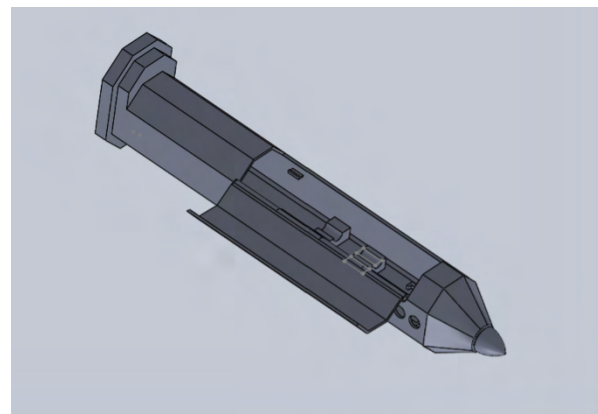


Figure 13: The CAD Assembly in SolidWorks of the final iteration of the base, casing, backing, overhead flap cover, and nib.

Rapid Prototyping via 3D Printing

Three prototypes were printed using FDM 3D printing. The first was printed in PLA, the second and final in ABS. The final prototype was a partial print of the inner base and pusher only, used for fit-testing internal components before committing to a full-enclosure print (Figure 14).

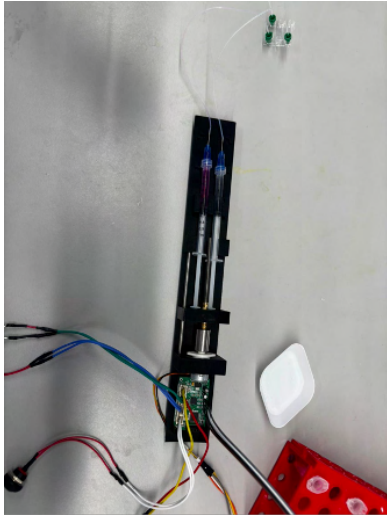


Figure 14: Prototype 3 (final iteration): 3D-printed inner base and pusher components during fit-testing of internal parts.

Table 7 in appendix I summarises the progression of the enclosure design across all iterations.

4.3 Overall Testing

Electrical Control Signal Testing

Oscilloscope measurements were taken during motor-control testing to verify stable switching behaviour under different actuation states. Figure 15 shows the PCB output during forward actuation at 17.66 kHz, while Figure 16 shows the corresponding reverse-actuation waveform. In both cases, stable square-wave switching was observed, indicating that the control circuit could operate well above the target stepping frequency of approximately 1.3 kHz.

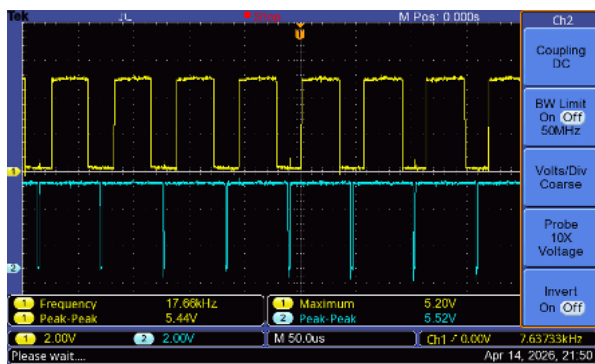


Figure 15: Oscilloscope trace of the PCB output during motor-control testing, showing stable square-wave switching during forward actuation at 17.66 kHz.

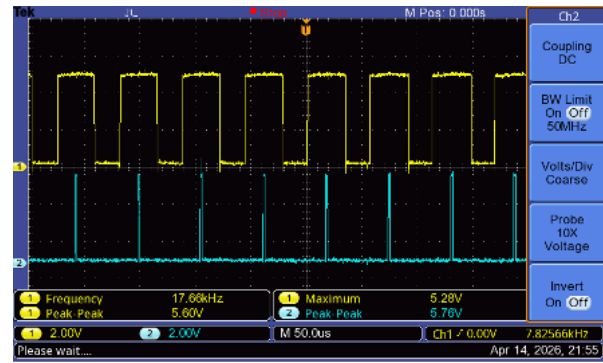


Figure 16: Oscilloscope trace of the PCB output during motor-control testing, showing stable square-wave switching during reverse actuation.

Mechanical Actuation Testing (Low-Load Condition)

The actuation mechanism was tested under low-load conditions using water-filled syringes. Under these conditions, the motor-driven lead screw system produced consistent linear displacement of the syringe plunger.

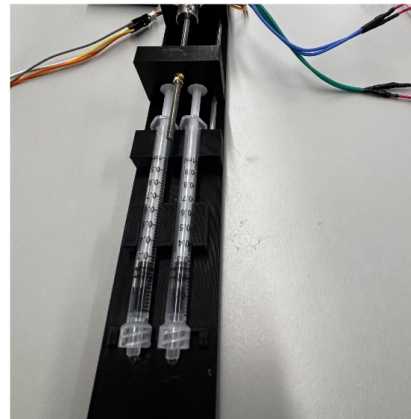


Figure 17: Low-load actuation test showing the syringe assembly and visible droplet formation at the outlet during motor-driven plunger displacement.

At the outlet controlled water was observed, confirming effective conversion of rotational motion into linear actuation. The system operated smoothly with minimal vibration, indicating stable mechanical performance and adequate alignment of moving components.

Fluidic and Integrated System Testing

System performance was evaluated using a 4% gelatin / 4% alginate hydrogel and the integrated microfluidic chip.

Under these conditions, the motor was unable to achieve extrusion and consistently stalled during

actuation. To isolate the cause, manual force was applied directly to the syringe plunger, resulting in successful extrusion through the microfluidic channels. The failure is therefore attributed to insufficient actuator force from the motor due to high resistances of the hydrogel in the microfluidic chip.

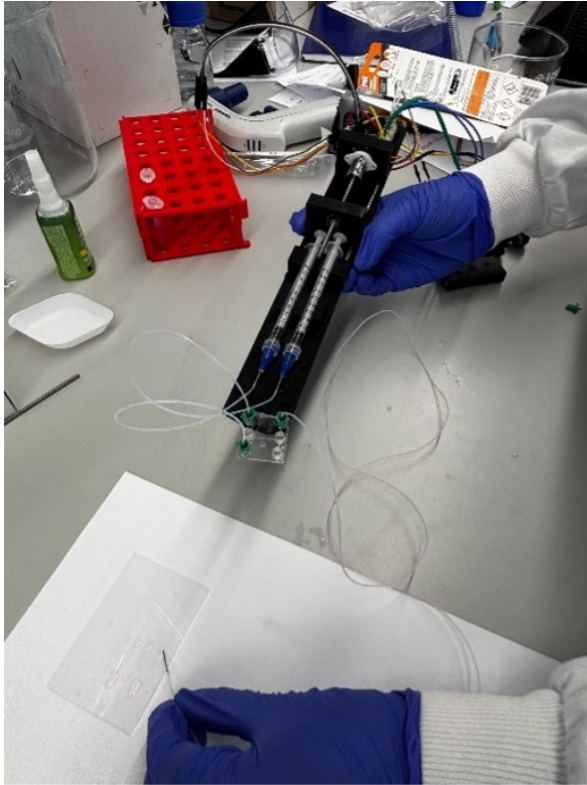


Figure 18: Experimental setup for fluidic and integrated system validation, showing the prototype connected to the microfluidic chip with gel emerging during manual plunger actuation after failure of motor-driven extrusion.

Start/Stop Response

The time delay between user input (button press) and initial motion of syringe pusher was measured at 1.40 ± 0.05 seconds ($n=5$). There is negligible delay between input and stopping of motor motion.

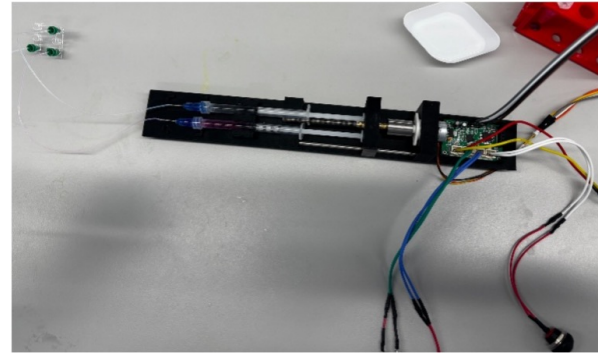


Figure 19: Assembled handheld bioprinter prototype used for system-level analysis, showing the integrated mechanical and electrical subsystems.

Questionnaire-Based User Evaluation

A total of 48 participants completed the questionnaire, providing a dataset to assess the perceived usability and clinical applicability of the handheld bioprinter.

Parameter	Mean $\pm$ SD	Key Finding
Ease of Use	4.67 ± 0.48	Highest rated; highly intuitive operation
Interface intuitiveness	4.62 ± 0.67	Effective simplified control system
Point-of-care suitability	4.56 ± 0.58	Strong support for compact, bedside use
Clinical potential	4.44 ± 0.74	High perceived ability to improve treatment
Adaptability to irregular geometries	4.35 ± 0.76	Supports key functional advantages
Ergonomics	4.19 ± 0.84	Lowest score; indicates area for refinement

Table 5: Summary of questionnaire responses assessing perceived usability and clinical relevance of the handheld bioprinter concept, including mean Likert scores ($\pm$ standard deviation) and associated key findings.

The distribution of responses and participant background information are presented in the Appendix J.

4.4 Future Steps

Future work will focus on the final system integration and validation of the handheld bioprinter, including completion of the remaining assembly steps

and integrated testing with the microfluidic subsystem to assess overall performance against design requirements.

Usability and ergonomic performance will also be evaluated through structured user studies, with comparative assessment against the original supervisor-provided model to assess potential improvements in handling, control, and user comfort during prolonged operation.

5 Discussion

5.1 Outline of Novelties Incorporated

On-board power and portability

A key feature of the present design is the integration of rechargeable on-board power within a compact handheld format. Unlike existing handheld bioprinting systems that rely on external power, pneumatic supply, or other bench-top hardware [25, 26, 27], the present device was designed as a self-contained unit incorporating all components within a single enclosure. This supports untethered operation and improves portability. The compact octagonal housing also supports single-handed operation and reduces rotational slippage.

Comparison with existing devices

Table 6 compares the present device with representative bioprinting systems reported in the literature. Relative to these devices, the main distinction of the present design lies in the combination of on-board rechargeable power, compact integration, and tool-free syringe access within a handheld format.

Feature	BioPen [25]	Skin Printer [26]	Open-Source [27]	Present Device
Power source	External pneumatic + UV	External benchtop	External supply	Rechargeable Li-ion (on-board)
Portable / untethered	No	No	No	Yes
Grip profile	Cylindrical (titanium)	Not handheld	Cylindrical (ABS)	Octagonal (ABS)
Syringe access	Integrated cartridge	Micro-fluidic cartridge	Full dis-assembly	Hinged top shell (tool-free)
Estimated cost	High (titanium, UV)	Moderate-high	Low	Low (3D-printed ABS)

Table 6: Comparison of the present device with existing handheld bioprinters across key portability and usability features.

5.2 Areas for Improvement

Electrical system limitations

The present electrical design remains limited by both control strategy and system validation. The system currently operates as an open-loop system, meaning that motor actuation is used as a proxy for extrusion without direct feedback on the delivered output. In extrusion-based bioprinting, process variations can lead to discrepancies between the intended actuation input and the actual delivered flow, while feedback-based control can improve precision and reproducibility [28,29].

Incorporating a closed-loop control strategy would therefore be an important next step towards improving extrusion precision and repeatability. In addition, although preliminary power measurements enabled a theoretical estimate of battery runtime, the electrical system has not yet been validated under extended operating conditions. Long-duration testing of battery endurance and supply stability would be required to assess the suitability of the device for sustained practical use.

Mechanical system limitations

The mechanical design was developed to support a compact and portable handheld bioprinter. However, the current ABS hinge mechanism introduces durability concerns, as repeated opening and closing may lead to wear overtime. Using a fastened metal hinge instead would improve the assembly's fatigue resistance.

Material selection remains another critical area for improvement. PLA was suitable for early prototyping due to its ease of fabrication and rapid iteration capabilities; however, its relatively low mechanical strength and limited thermal resistance reduce its suitability for functional use [30]. Consequently, the final prototype was fabricated using ABS to enhance structural robustness. However, ABS is generally not compatible with standard autoclave sterilization (typically 121°C or 134°C), as its glass transition temperature is approximately 105°C, potentially limiting its deployment in clinical settings [22]. Future iterations may therefore benefit from high-performance, autoclavable engineering thermoplastics such as polysulfone (PSU) or polyphenylsulfone (PPSU), which are more suitable for repeated sterilisation and for more clinically applicable device [21,31].

Testing Limitations

A key limitation of the testing was that the current actuator configuration did not yet provide sufficient force for consistent hydrogel extrusion under representative loading conditions. This is due to the motor not providing enough torque to overcome the combined resistance from the syringe plunger, microfluidic system and the frictional drag between the plunger slab and the base. Manual plunger actuation confirmed the gel delivery through the microfluidic pathway was achievable; therefore using a more powerful motor would be the next step.

5.3 Clinical Consultation and Design Implications

The clinical consultation provided a useful framework for assessing how effectively the prototype addressed practical bedside requirements. Several priorities identified by the consultant doctor were

reflected in the final design, including portability, simplified operation and support for user-guided deposition over irregular wound geometries. The self-contained architecture, two button interface and pen-like form therefore aligned with clinically motivated design requirements rather than purely technical considerations.

However, some clinically relevant requirements that the consultant identified remain to be fully addressed. Sterility considerations were not comprehensively resolved beyond enabling internal access for cleaning and maintenance, as well as workflow integration not being validated in a clinical environment. These factors represent important considerations for future development and clinical translation.

5.4 User Evaluation Analysis

The questionnaire results suggest positive perceived usability and clinical relevance of the device. However, these findings should be interpreted cautiously because evaluation was based on video demonstration rather than direct hands-on use. The results therefore support design acceptability at a conceptual level, but do not yet provide strong evidence of real-world ergonomic or workflow performance. High mean scores for ease of use (4.67 ± 0.48) and interface intuitiveness (4.62 ± 0.67) indicate the simplified two-button control system was generally perceived as easy to understand. Point-of-care suitability (4.56 ± 0.58) and clinical potential (4.44 ± 0.74) were also rated positively, suggesting alignment between the device concept, and identified clinical needs. The rating for adaptability to irregular geometries (4.35 ± 0.76) is consistent with the intended advantage of handheld, in situ deposition over prefabricated constructs. Ergonomics received the lowest mean score (4.19 ± 0.84), indicating that the increased device size required for internal component integration may affect comfort during prolonged handling. This highlights a key design trade-off between portability and usability. Although based on visual demonstration rather than physical interaction, responses suggest favourable user perception whilst identifying areas for further refinement.

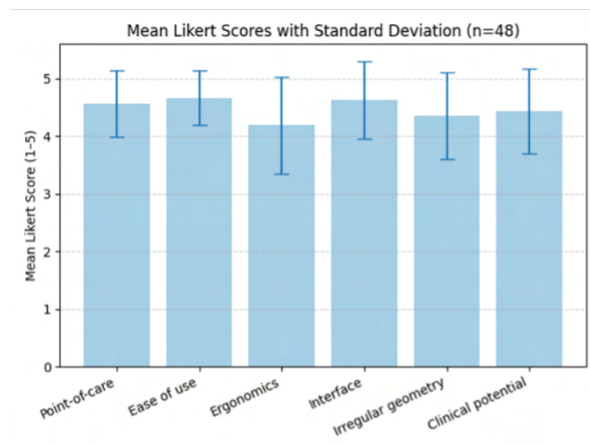


Figure 20: Mean Likert Scores ($\pm$ SD, $n = 48$) for measures of perceived usability and clinical relevance of the handheld bioprinter concept.

Comparison between participant groups

Independent t-tests comparing participant groups (Medicine/Healthcare; $n = 12$; Biomedical Engineering; $n = 16$) revealed that medical participants rated point-of-care suitability (4.83 vs 4.31, $p = 0.010$) and clinical potential (4.67 vs 4.00, $p = 0.024$) much higher than bioengineering participants, with a similar trend for adaptability to irregular geometries.

Bioengineering participants reported slightly lower ergonomics scores, though this was not statistically significant, while ease of use was consistent across groups.

These findings suggest that clinically aligned participants emphasize application potential, whereas engineering participants adopt a more critical perspective on technical implementation, reflecting the balance between clinical need and engineering constraints.

Participant Background Distribution (n=48)

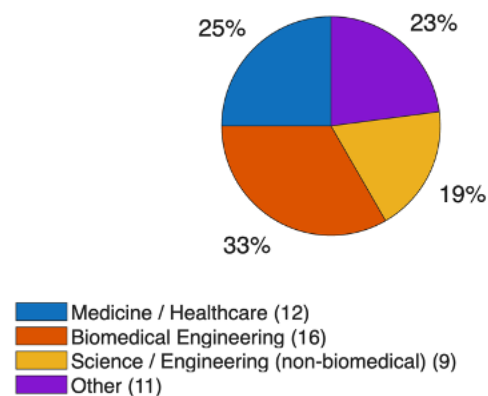


Figure 21: Distribution of participant backgrounds in the questionnaire cohort ($n = 48$), showing the relative proportions of biomedical engineering, medicine/healthcare, science/engineering (non-biomedical), and other backgrounds.

6 Conclusion

This project developed and evaluated the electrical and mechanical subsystems of a portable handheld bioprinter platform for potential point-of-care wound-healing applications. Breadboard validation and iterative PCB development demonstrated reliable programming, motor actuation, and standalone electrical operation, while the mechanical design established a compact enclosure concept for handheld use. Together, these results provide a proof of concept for a self-contained bioprinter platform, though full system integration and extrusion validation under high resistance conditions remain to be addressed for clinical use.

7 Acknowledgements

We would like to express our gratitude towards Ms Rita Correia, Dr. Nicola Contessi Negrini and Dr. Adam Celiz for their continued support, advice and encouragement. We would also like to extend a special thanks to all the technicians that have provided us with invaluable project advice.

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APPENDIX

A) Nomenclature

Full-Forms:

23G: 23 Gauge Needle

ABS: Acrylonitrile Butadiene Styrene

CAD: Computer Aided Design

DIP: Dual In-line Package

FDM: Fused Deposition Modelling

ISP: In-System Programming

LED: Light Emitting Diode

MCU: Microcontroller Unit

PCB: Printed Circuit Board

PLA: Polylactic Acid

SMD: Surface Mount Device

UI: User Interface

UV: Ultraviolet

Biological:

Angiogenesis: Angiogenesis is the growth of blood vessels from the existing vasculature.

Arduino: Arduino is an open-source platform used for building electronics projects.

Bioink: Bio ink is defined as a suitable composition of cells along with biologically active components and biomaterials, processed for use in bioprinting.

Bioprinting: Bioprinting is a technology where bioinks and biomaterials, mixed with cells, are 3D printed, often to construct living tissue models.

Breadboard: A breadboard is a rectangular, white board filled with small, interconnected holes that enable the assembly of electronic circuits.

Buck Converter: A buck converter or step-down converter is a DC-to-DC converter which decreases voltage, while increasing current, from its input (supply) to its output (load).

Crystal Oscillator: A crystal oscillator is defined as a device that utilizes a quartz crystal to generate stable electrical signals by converting electrical energy to mechanical energy and vice versa.

Extracellular Matrix (ECM): A large network of proteins and other molecules that surround, support, and give structure to cells and tissues in the body.

Firmware: Firmware is a small software that is written to Hardware devices in non-volatile memory.

Flow rate: Volume of fluid which passes per unit time.

Haemostasis: Body's way of stopping bleeding and repairing the wound after an injury.

Hydrogel: Hydrogel is a hydrophilic polymer synthesized from natural or synthetic materials with a three-dimensional network structure.

In-situ: 'in the place' or 'on site' (Latin).

Interference Fit: An interference fit, also known as a press fit or friction fit, is a form of fastening between two tight fitting mating parts that produces a joint.

Lead Screw Pitch: The pitch of a lead screw is essentially the distance between threads, measured from the top of one screw thread to the top of the next screw thread.

Micro-stepping: Microstepping is a method of controlling stepper motors, typically used to

achieve higher resolution or smoother motion at low speeds.

Non-Blocking Timers: Arduino Library which does the following: Unlike traditional timers, this library allows the creation of timers that operate independently, allowing the program to continue execution while the timer counts down in the background.

Potentiometer: The potentiometer is an instrument used to measure the unknown voltage by comparing it with the known voltage.

Proliferation: Proliferation is the growth of tissue cells.

Scaffold: A biological or artificial structure that provides support and maintains tissue contour.

Stepper Motor: A stepper motor, also known as a step motor or stepping motor, is a brushless DC electric motor that rotates in a series of small and discrete angular steps.

Vascularisation: The organic process whereby body tissue develops capillaries.

B) Circuit Diagrams and Schematics

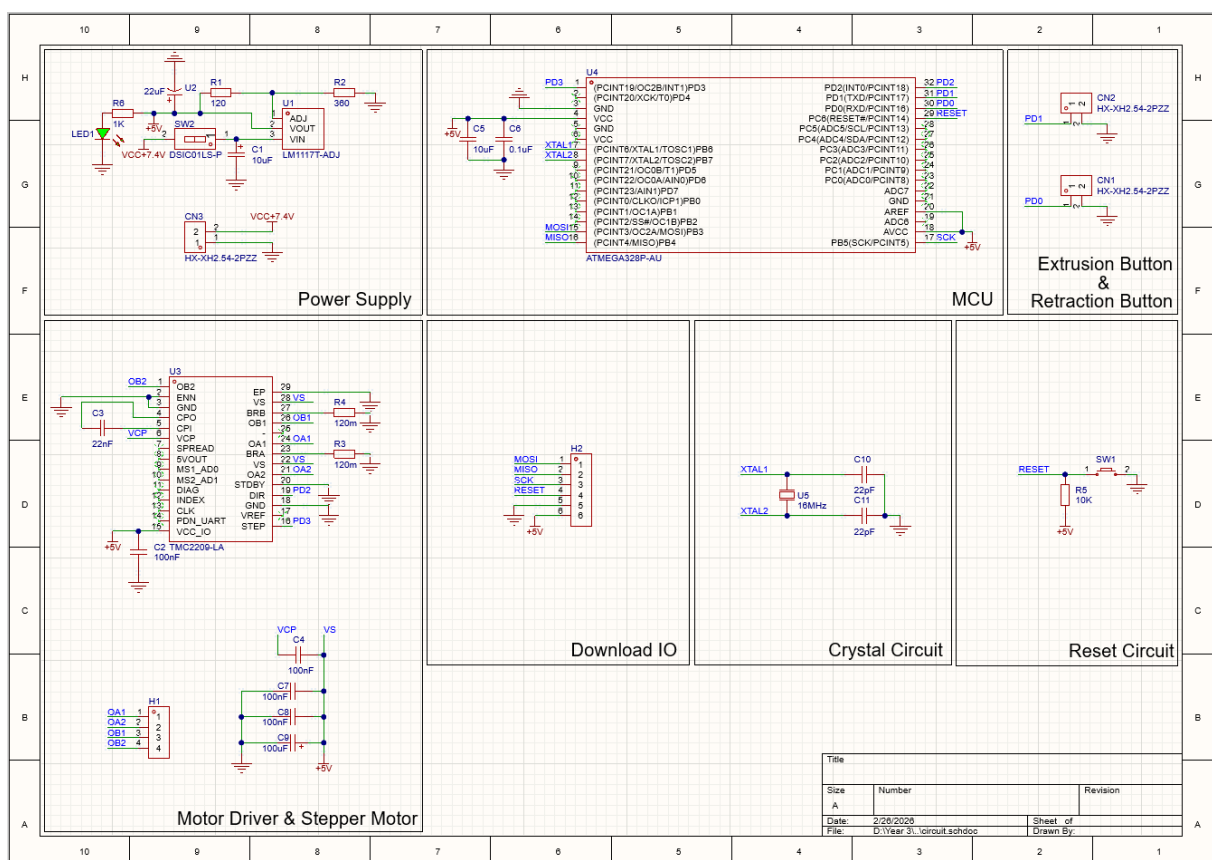


Figure 22: First version of schematic sheet of the circuit

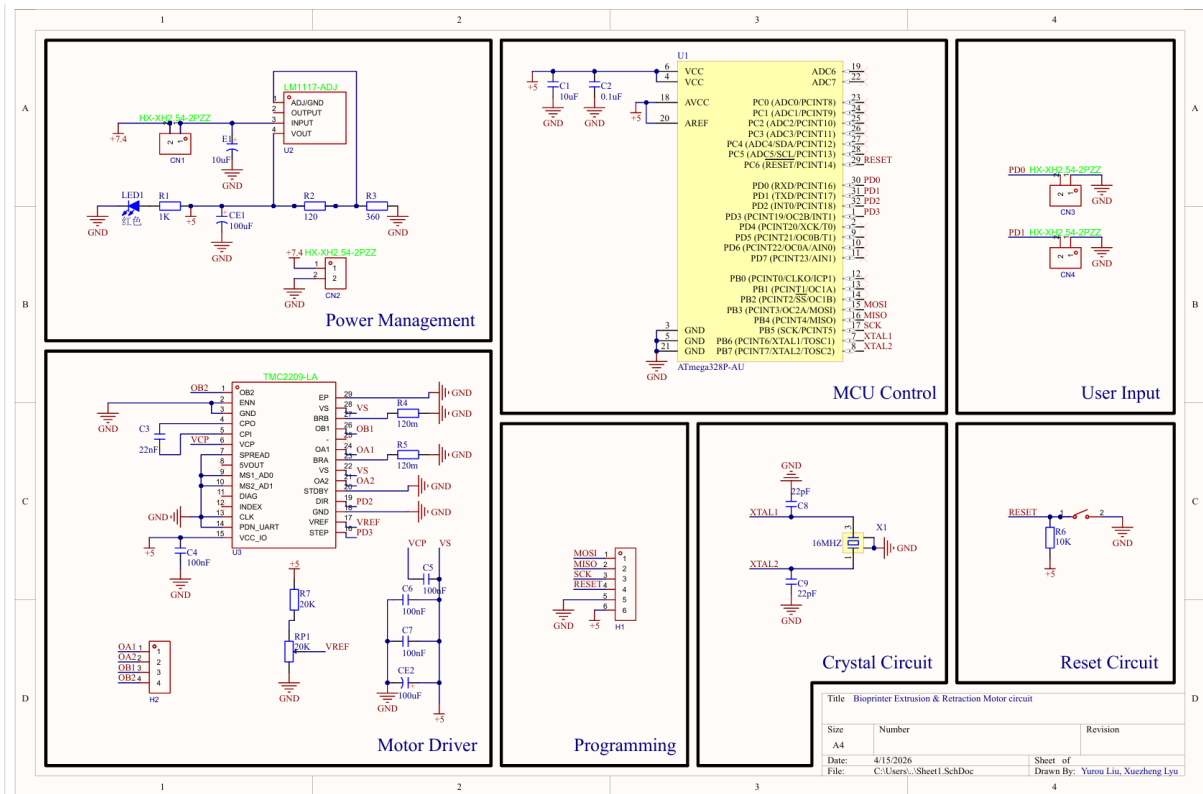


Figure 23: Second version of schematic sheet of the circuit

C) PCB Diagrams

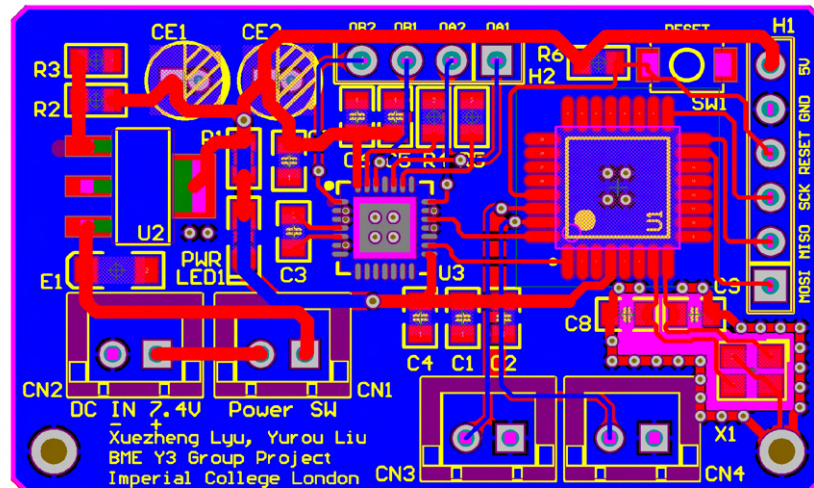


Figure 24: First version of PCB layout

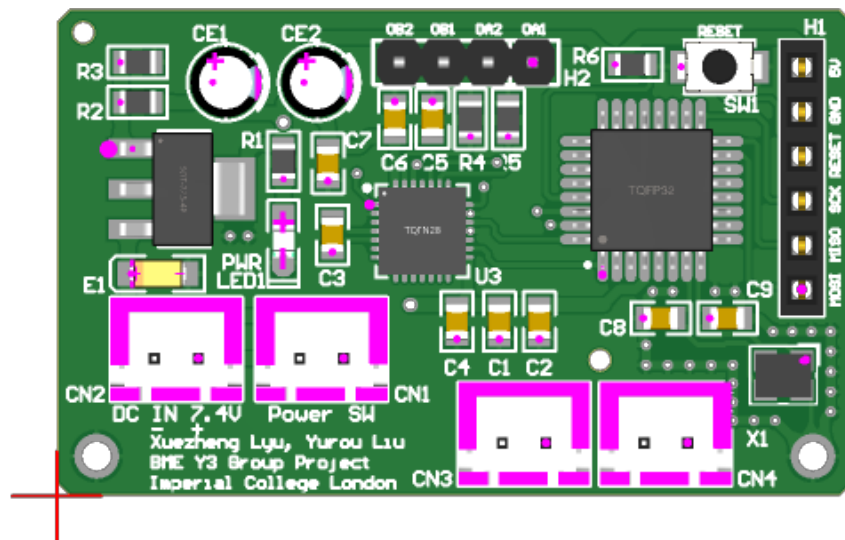


Figure 25: First version of 3D top view of the PCB

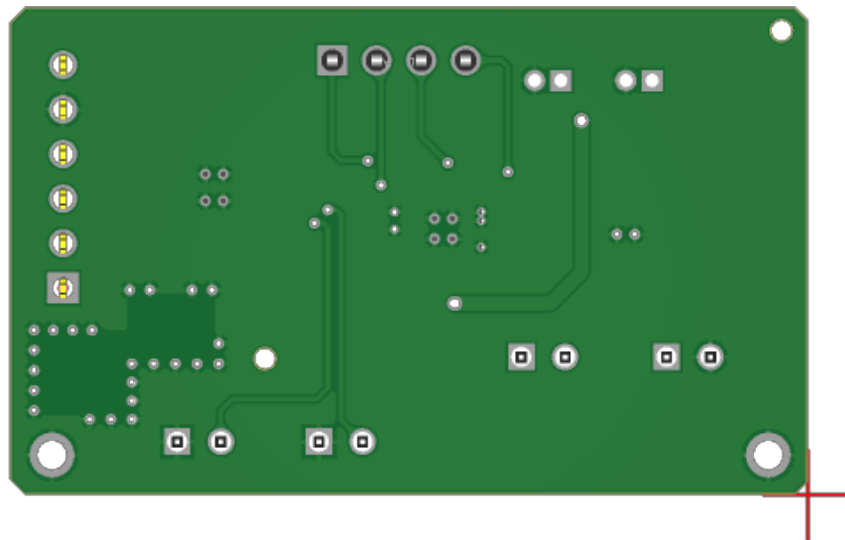


Figure 26: First version of 3D bottom view of the PCB

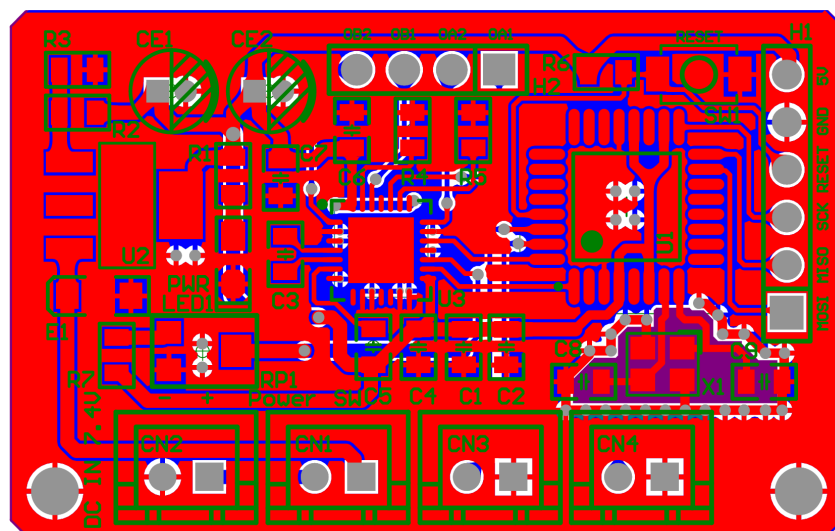


Figure 27: Second version of PCB layout

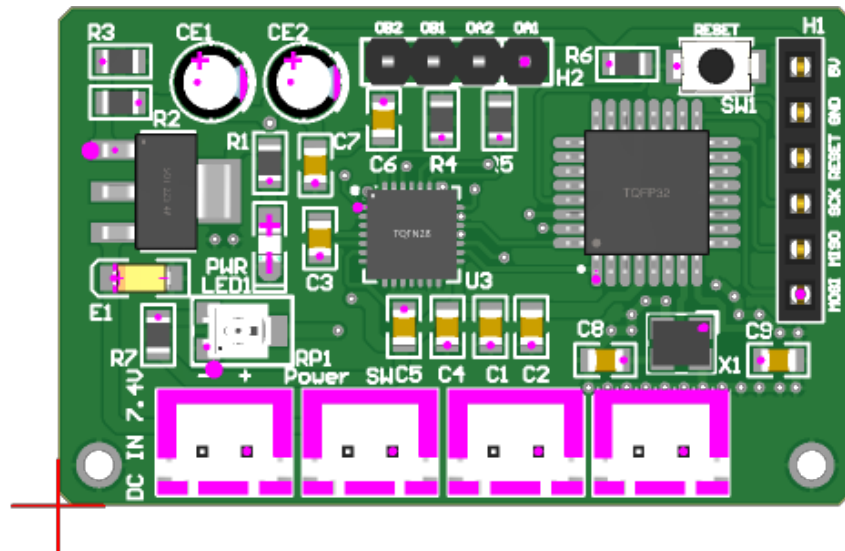


Figure 28: Second version of 3D top view of the PCB

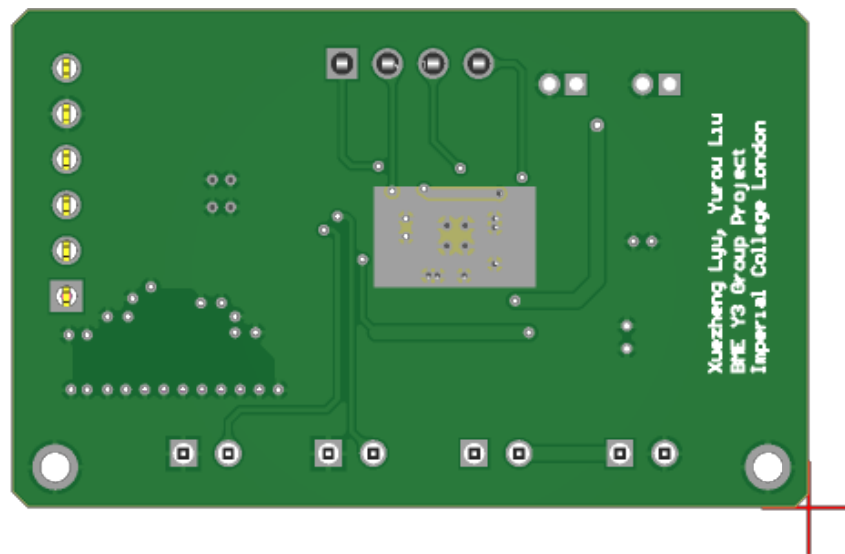


Figure 29: Second version of 3D bottom view of the PCB

D) Lithium Discharge Formula

Theoretical energy:

$$Q = \frac{E}{V} = \frac{0.75\text{Wh}}{3.7\text{V}} \approx 0.2027\text{Ah} \quad (1)$$

Theoretical discharge rate:

$$C = \frac{I}{Q} = \frac{0.18\text{A}}{0.2027\text{Ah}} \approx 0.888\text{C} \quad (2)$$

Theoretical working time:

$$T = \frac{1}{0.888\text{C}} \times 60 \approx 67.6 \text{ min} \quad (3)$$

E) Calculation Logic for the Arduino Code

Total Syringe Cross-sectional Area:

$$A_{\text{single}} = \frac{\text{Volume}}{\text{Stroke}} = \frac{1000 \mu\text{L}}{60 \text{ mm}} = 16.667 \text{ mm}^2 \quad (4)$$

$$A_{\text{total}} = A_{\text{single}} \times 2 \text{ (Dual Syringes)} = 33.333 \text{ mm}^2 \quad (5)$$

Target Linear Velocity:

Target Flow Rate: $Q = 1225 \mu\text{L/h}$.

$$v = \frac{Q}{A_{\text{total}}} = \frac{1225 \text{ mm}^3/\text{h}}{33.333 \text{ mm}^2} = 36.75 \text{ mm/h} \quad (6)$$

Motor Output Shaft Rotational Speed:

Assuming Lead: $L = 0.5 \text{ mm/rev}$.

$$\omega = \frac{v}{L} = \frac{36.75 \text{ mm/h}}{0.5 \text{ mm/rev}} = 73.5 \text{ rev/h} \quad (7)$$

Pulse Frequency:

$$\text{Total Pulses/Revolution} = \text{Steps/Rev} \times \text{Microstepping} = 2000 \times 32 = 64,000 \text{ pulses/rev} \quad (8)$$

$$f(\text{Hz}) = \frac{\omega \times 64,000}{3600 \text{ (seconds/hour)}} = \frac{73.5 \times 64,000}{3600} = 1306.67 \text{ Hz} \quad (9)$$

Code Execution Parameter:

Pulse Period: $T = 1/f = 765.3 \mu\text{s}$.

$$T_{\text{half}} = \frac{T}{2} = 382.6 \mu\text{s} \approx 383 \mu\text{s} \quad (10)$$

F) Arduino Code

```

1  // ===== TMC2209 STEP/DIR + Dual Button Control =====
2  // Logic: Long press (1s) to Start; Short click during operation to Stop.
3  // Target Flow Rate: 1225 uL/h (Combined for 2 syringes)
4
5  const int stepPin = 3; // STEP pin
6  const int dirPin  = 2; // DIR pin
7  const int BTN_A   = 0; // Forward control (PD0)
8  const int BTN_B   = 1; // Reverse control (PD1)
9
10 // --- Physical Parameters ---
11 const float TARGET_FLOW_UL_PER_H = 1225.0f;
12 const float SYRINGE_VOL_UL       = 1000.0f; // 1mL capacity
13 const float SYRINGE_STROKE_MM    = 60.0f;   // 60mm usable stroke
14 const int   NUM_SYRINGES         = 2;
15 const float SCREW_LEAD_MM        = 0.5f;     // Estimated M3 lead screw
16 const long  STEPS_PER_REV        = 2000;     // 20 steps * 100 gear ratio
17 const int   MICROSTEP            = 32;       // TMC2209 MS setting
18
19 unsigned long pulseIntervalUs = 0; // Calculated dynamically
20
21 enum RunState { STOPPED, FORWARD, REVERSE };
22 RunState state = STOPPED;
23
24 struct Button {
25     int pin;
26     bool lastState;
27     unsigned long pressMs;
28     bool triggered;
29 };
30
31 Button btnA = {BTN_A, HIGH, 0, false};
32 Button btnB = {BTN_B, HIGH, 0, false};
33
34 void calculateTiming() {
35     float totalArea = (SYRINGE_VOL_UL / SYRINGE_STROKE_MM) * NUM_SYRINGES;
36     float volPerRev = totalArea * SCREW_LEAD_MM;
37     float revPerHour = TARGET_FLOW_UL_PER_H / volPerRev;

```

Figure 30: Arduino code screenshot (Part 1)

```
38
39     float freqHz = (revPerHour * STEPS_PER_REV * MICROSTEP) / 3600.0f;
40     pulseIntervalUs = (unsigned long)(500000.0f / freqHz);
41 }
42
43 void handleButton(Button &b, RunState targetDir) {
44     bool reading = digitalRead(b.pin);
45     unsigned long now = millis();
46
47     if (reading == LOW && b.lastState == HIGH) { // Press detected
48         b.pressMs = now;
49         b.triggered = false;
50     }
51     else if (reading == LOW && !b.triggered) { // Holding
52         if (now - b.pressMs >= 1000) {
53             state = targetDir;
54             b.triggered = true;
55         }
56     }
57     else if (reading == HIGH && b.lastState == LOW) { // Release
58         if (!b.triggered && state == targetDir) state = STOPPED;
59     }
60     b.lastState = reading;
61 }
62
63 void motorPulse() {
64     static unsigned long lastUs = 0;
65     static bool level = LOW;
66
67     if (state == STOPPED) {
68         digitalWrite(stepPin, LOW);
69         return;
70     }
71
72     digitalWrite(dirPin, (state == FORWARD) ? HIGH : LOW);
73
74     if (micros() - lastUs >= pulseIntervalUs) {
75         level = !level;
76         digitalWrite(stepPin, level);
77         lastUs = micros();
78     }
79 }
80
81 void setup() {
82     pinMode(stepPin, OUTPUT);
83     pinMode(dirPin, OUTPUT);
84     pinMode(BTN_A, INPUT_PULLUP);
85     pinMode(BTN_B, INPUT_PULLUP);
86
87     calculateTiming(); // Initial calculation
88 }
89
90 void loop() {
91     handleButton(btnA, FORWARD);
92     handleButton(btnB, REVERSE);
93     motorPulse();
94 }
```

Figure 31: Arduino code screenshot (Part 2)

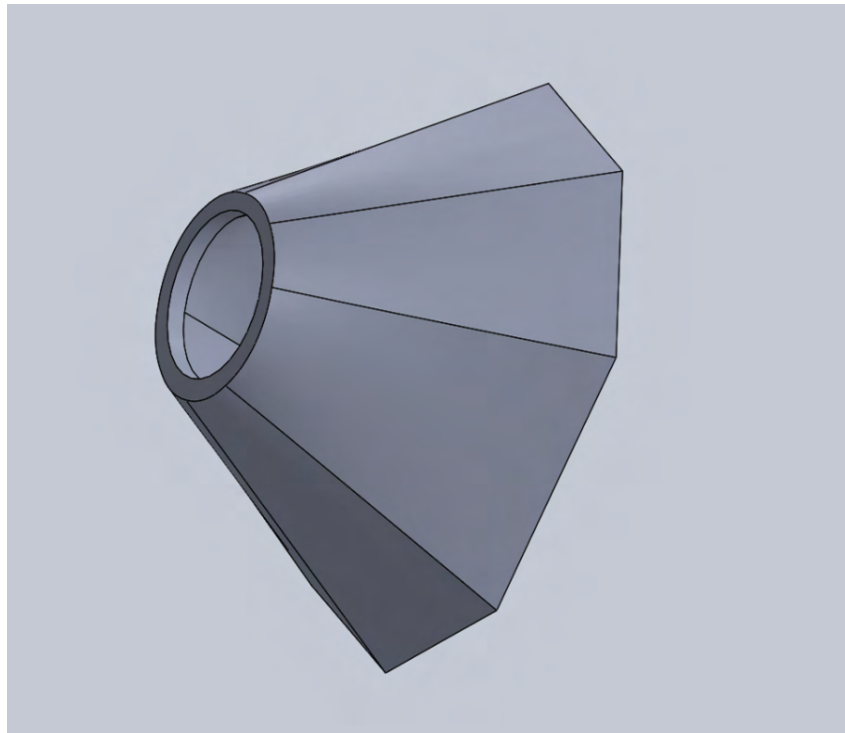
G) SolidWorks CAD Separate Components

Figure 32: Final Iteration Nib

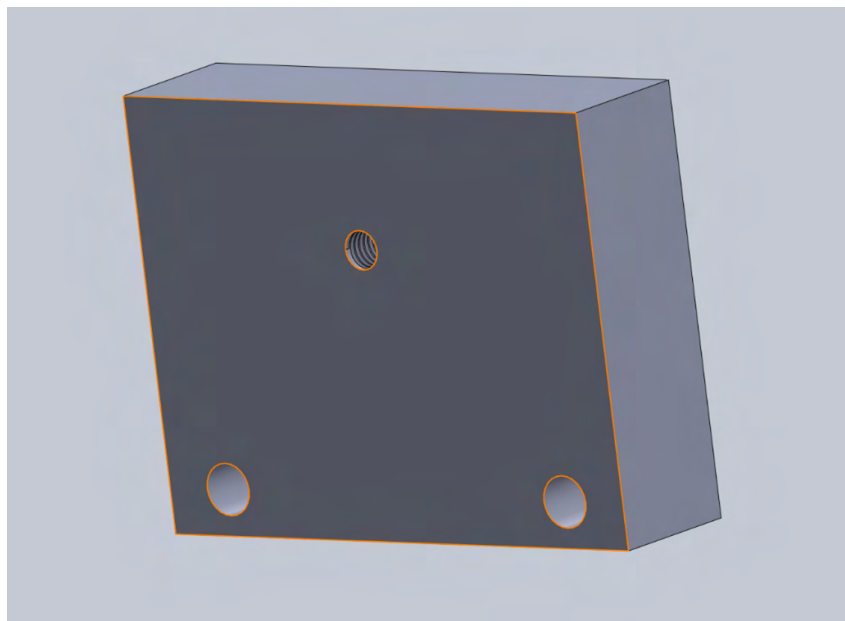


Figure 33: Final Iteration Slab

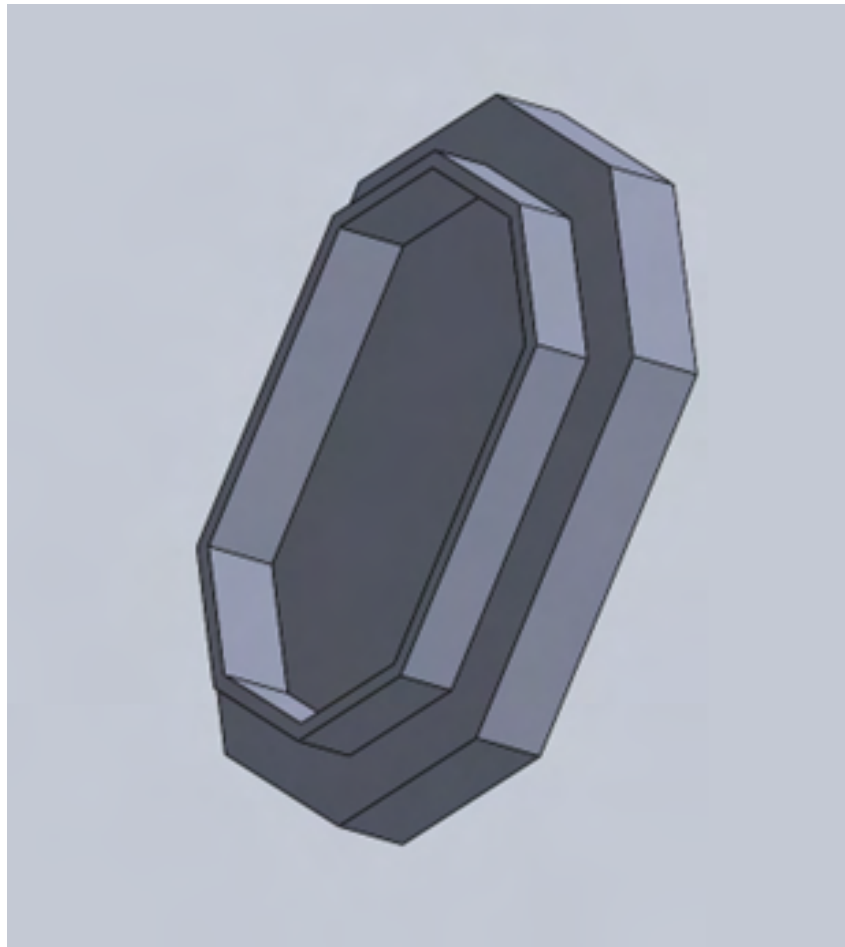


Figure 34: Final Iteration Back Cover

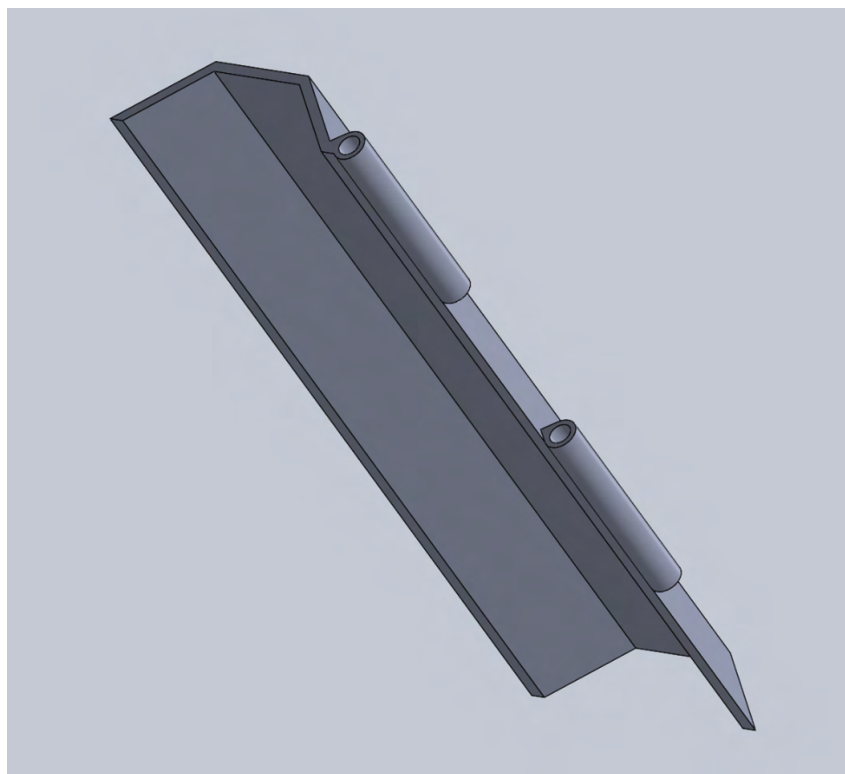


Figure 35: Final Iteration Flap Cover

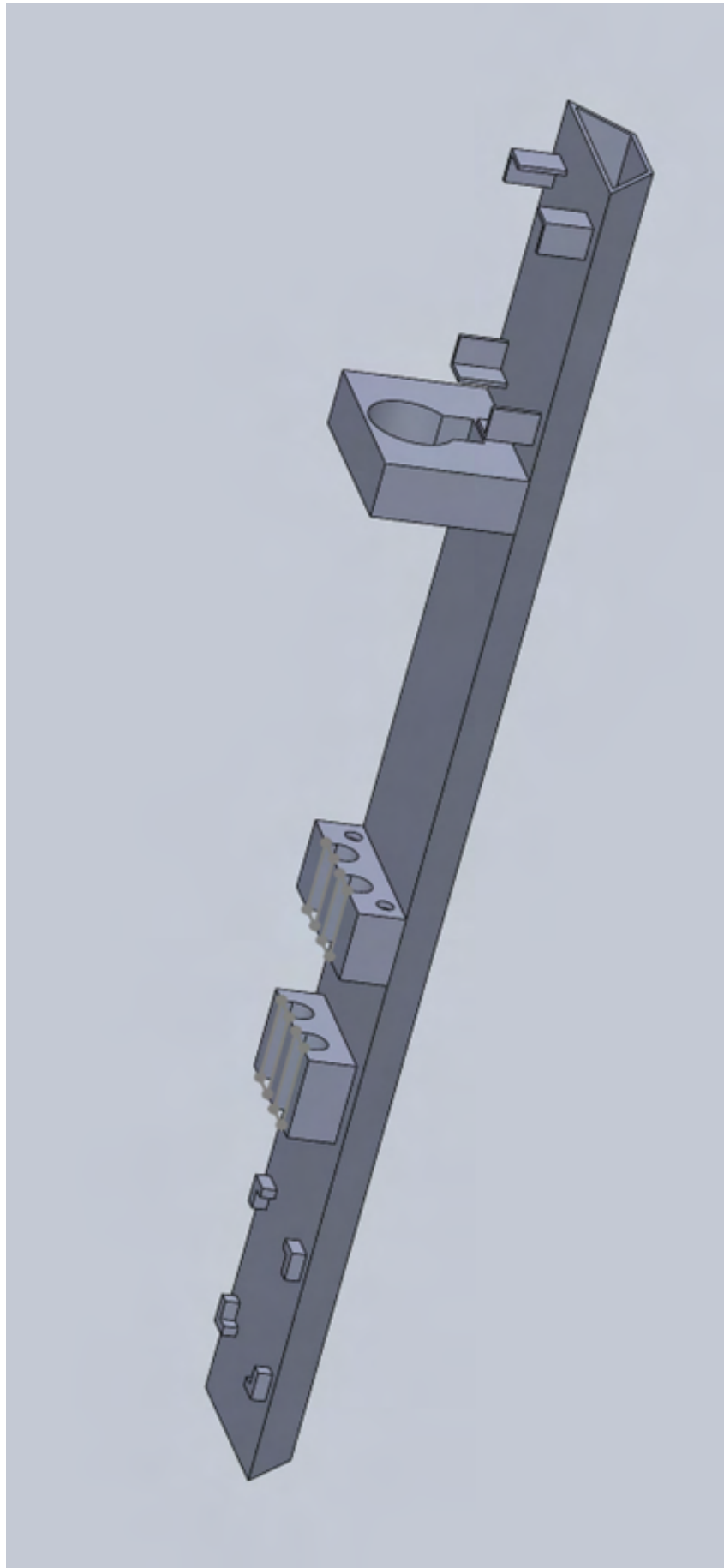


Figure 36: Final Iteration Base

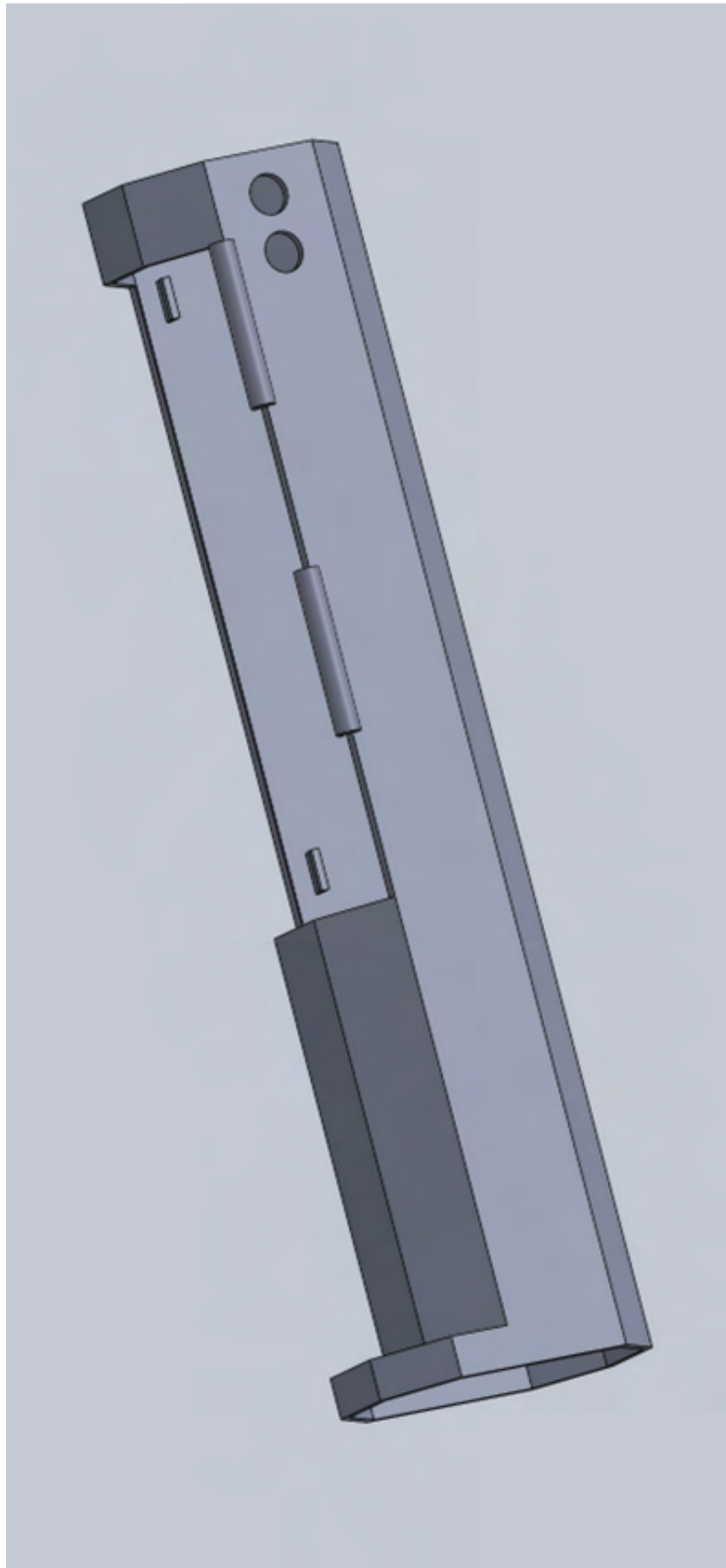


Figure 37: Final Iteration Cover

H) Technical Drawings

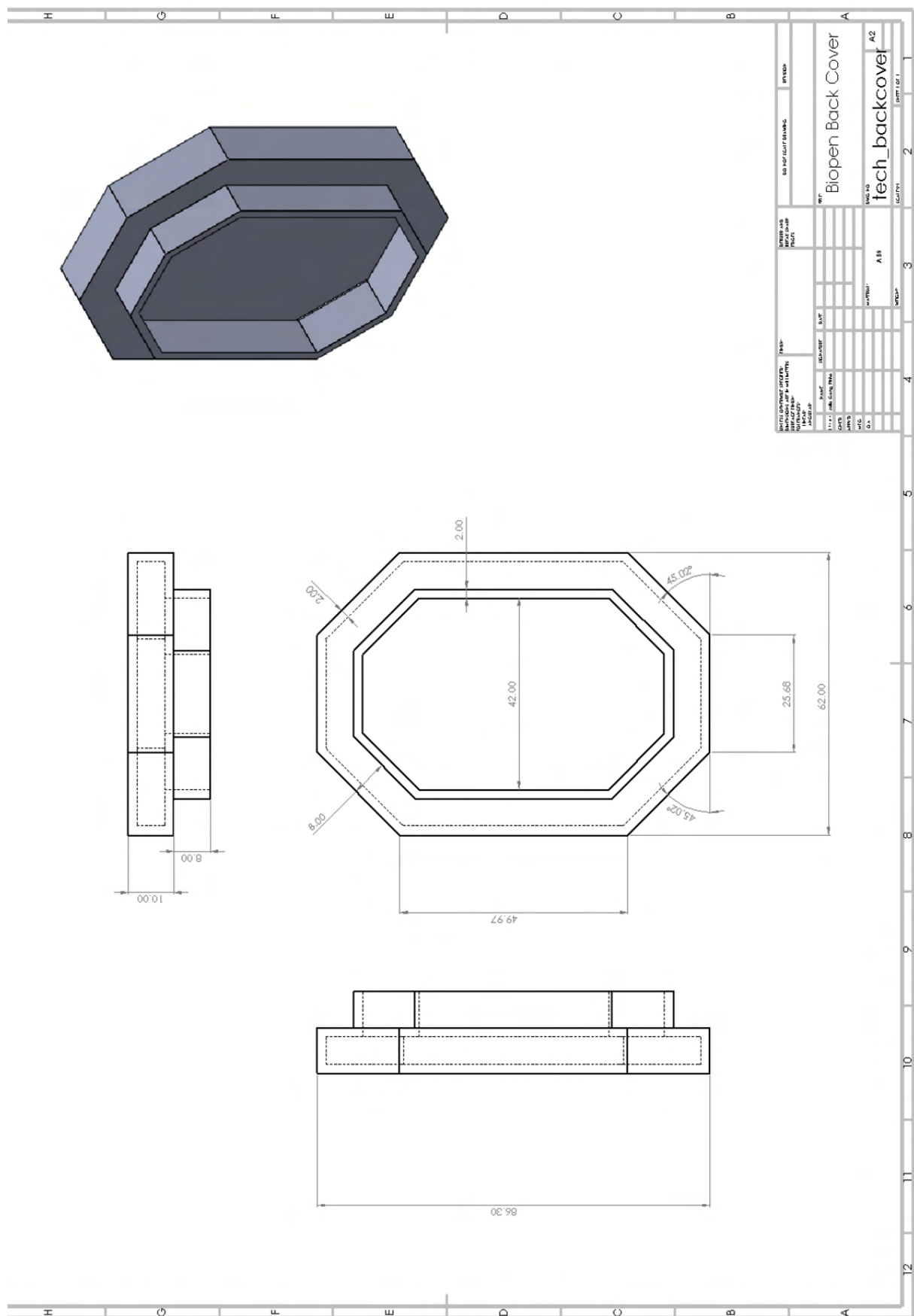


Figure 38: Final Iteration Slab

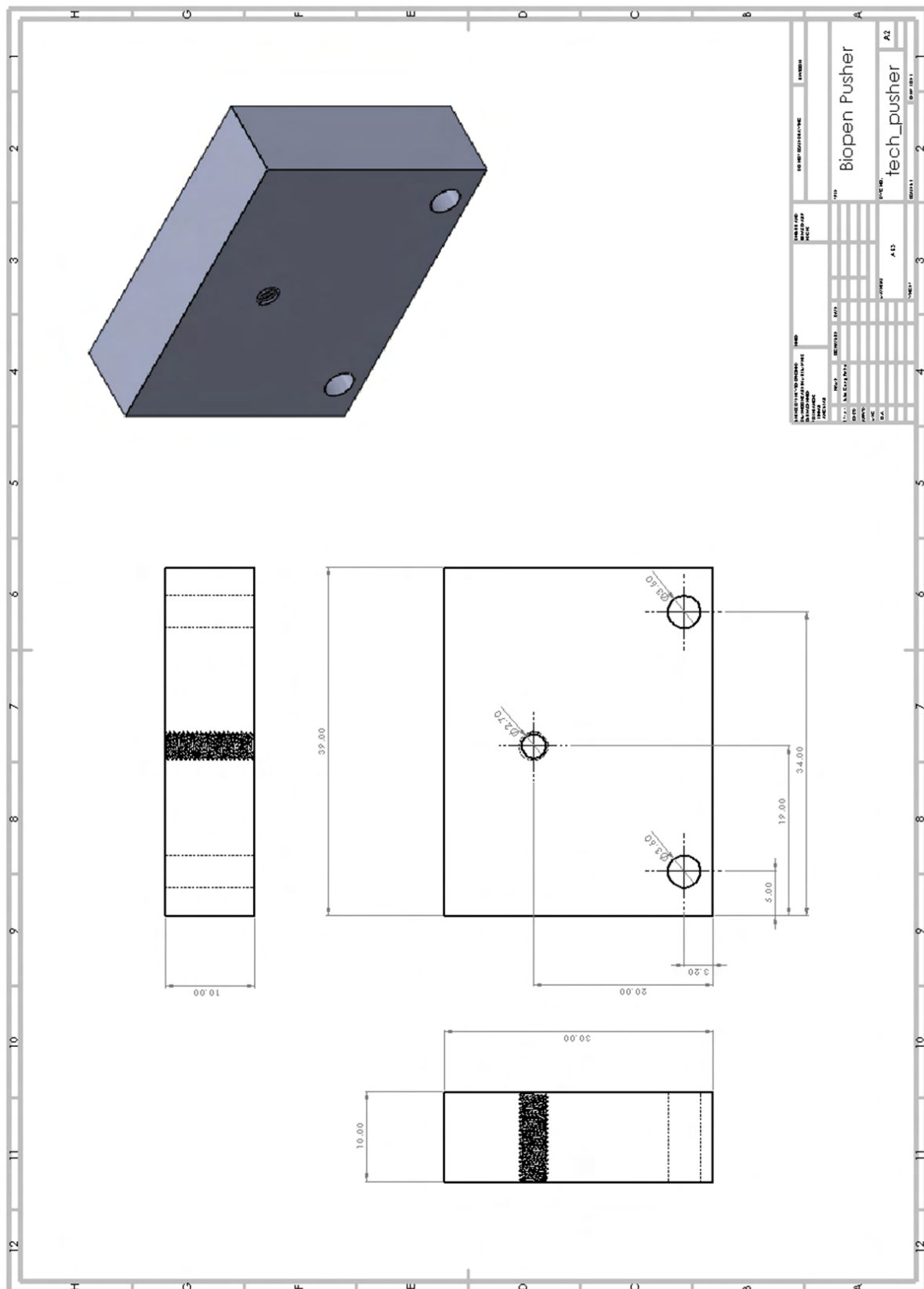


Figure 39: Final Iteration Pusher

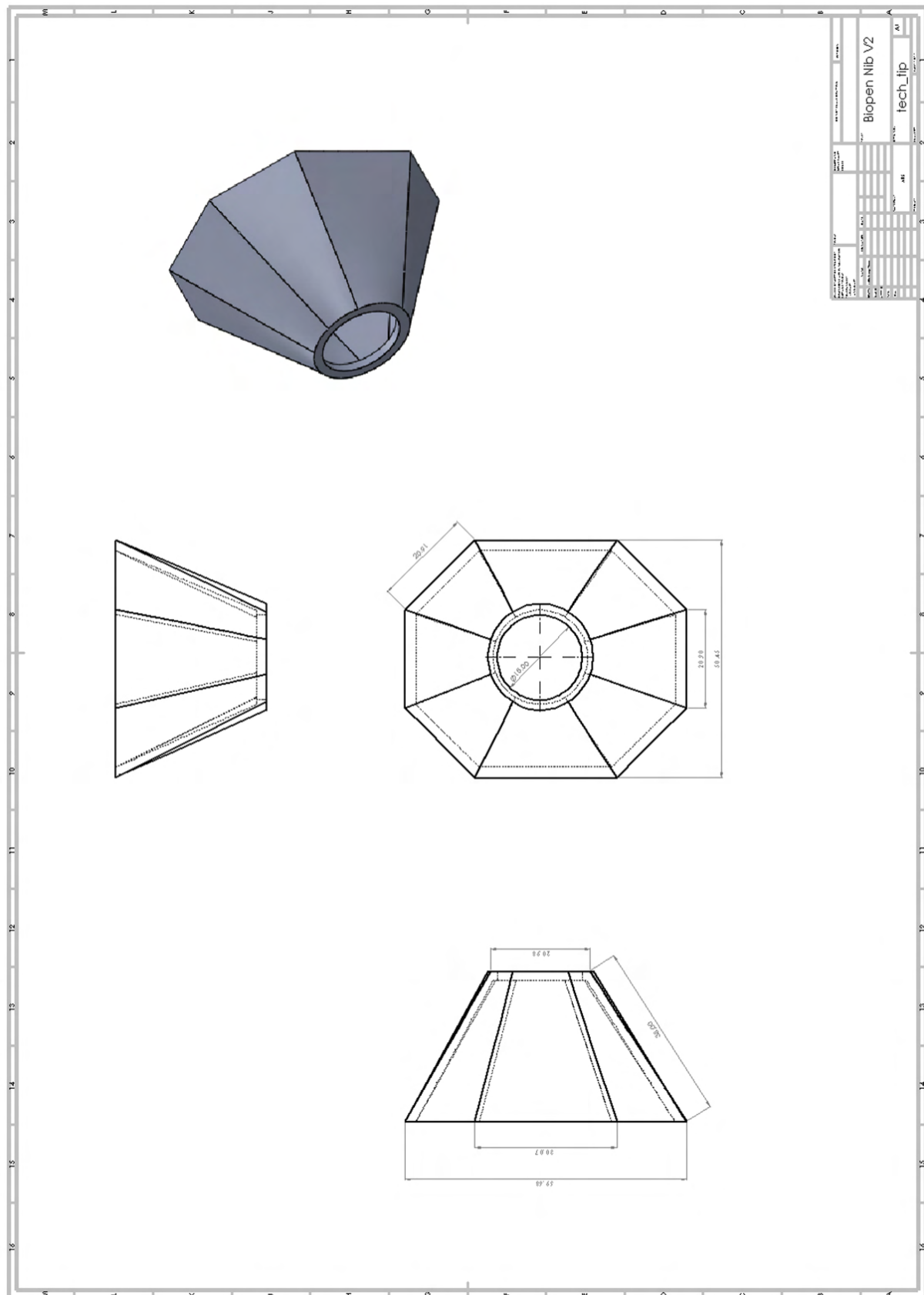
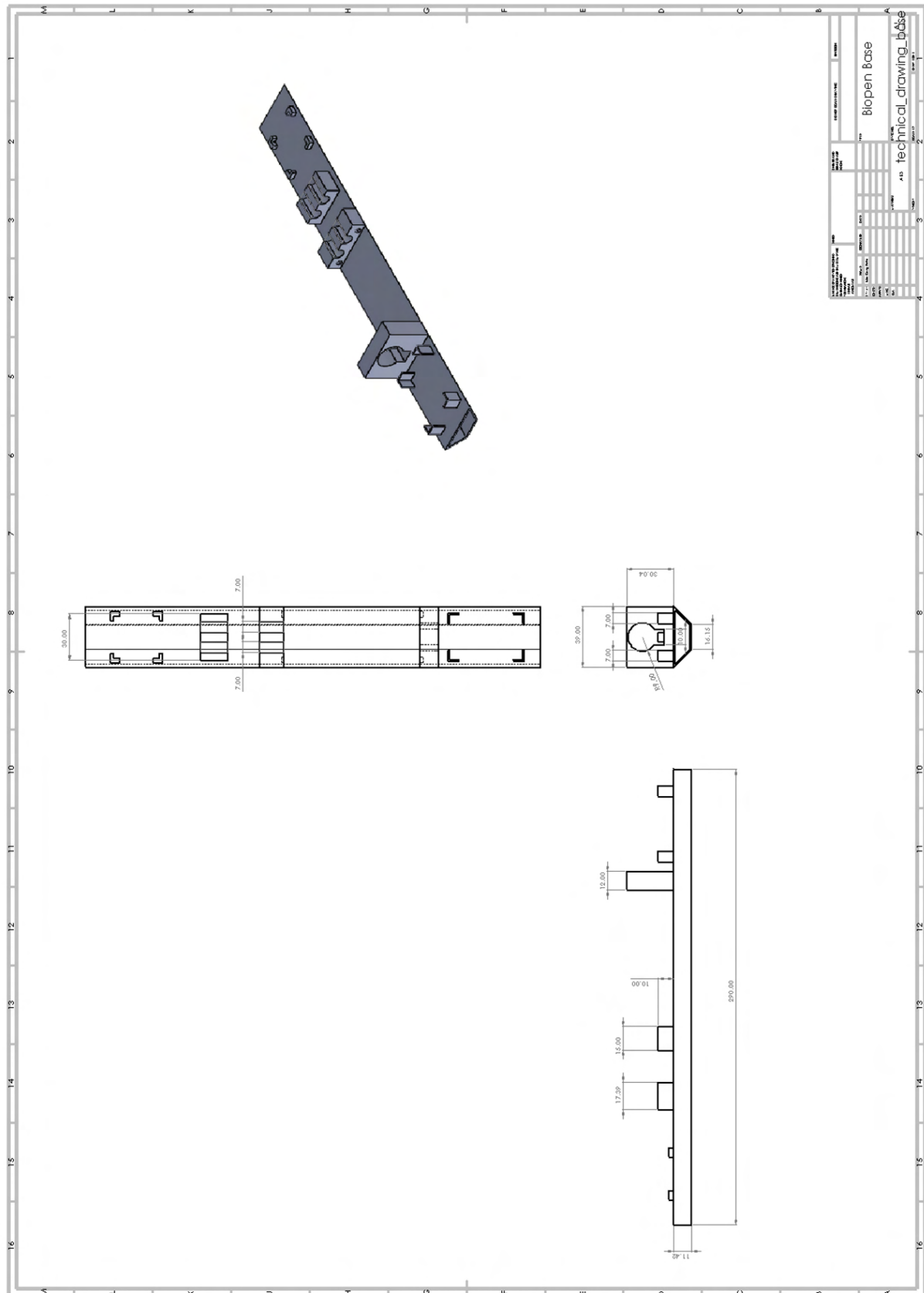


Figure 40: Final Iteration Nib

*Figure 41: Final Iteration Base*

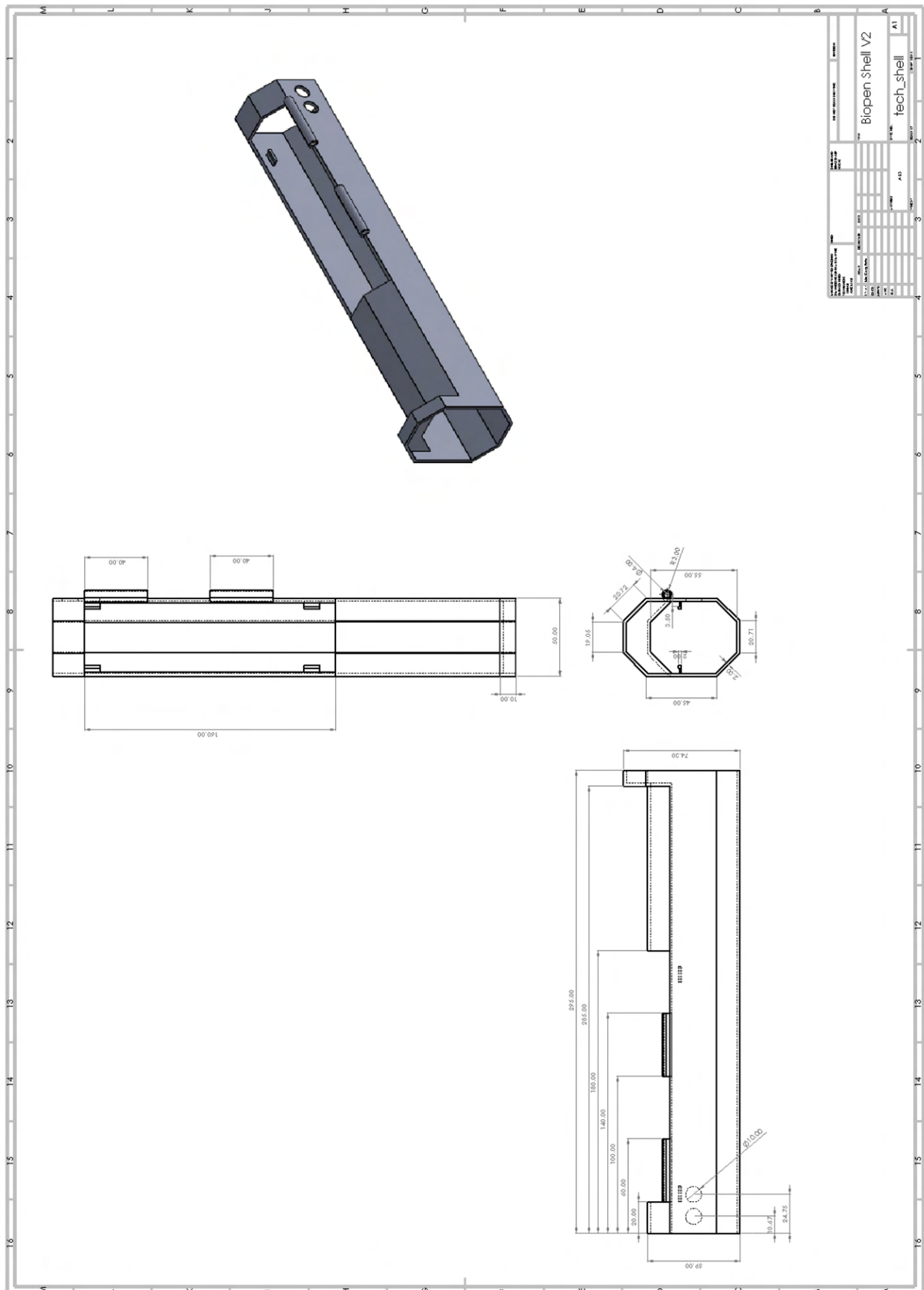


Figure 42: Final Iteration Casing

I) Summary of Enclosure Design Iterations

Iteration	Problem Identified	Design Change	Outcome
1.0	Cylindrical pen shapes allow rotational slippage; single-body shells prevent easy syringe access	Octagonal cross-section; 3-part split (top shell with hinge, bottom shell, inner base)	Improved grip via flat facets; modular assembly enabling syringe cartridge replacement
2.0	Battery larger than allocated octagonal cross-section; widening octagon would enlarge grip excessively; additional features to be added	Increased pen height (cross-sectional height, 46 to 55 mm); outer-shell re-profiled; added extrusion tip, removable battery backing (click-in/out); solid button model (position placeholder)	Battery accommodated without altering grip diameter; functional features for pitch demo; button position defined but no clearance for physical switch
Final	Battery still insufficiently mounted; button model blocks switch insertion; hinge too tight; nib size mismatch	Raised rear profile; modular pen nib; button clearance cutout; enlarged hinge + backing resized	Improved battery seating; all physical components accommodated; smoother hinge action; print-ready design

Table 7: Summary of enclosure design iterations: problems identified, design changes implemented, and outcomes achieved.

J) Questionnaire response analysis

You are invited to take part in a short survey evaluating the **design of a handheld bioprinter** developed as part of our 3rd Year Bioengineering Group Project. You will be shown a short demonstration video of a CAD model of the device. Please answer the following questions based on your perception of the **design, usability, and potential clinical application**.

All responses are anonymous and participation is voluntary. By completing this questionnaire, you consent to your responses being used for academic purposes within this project. Thank you for your participation.

1. What is your background? *

☐ Biomedical Engineering
☐ Medicine / Healthcare
☐ Science / Engineering (non-biomedical)
☐ Other

2. How familiar are you with wound care or regenerative medicine? *

	1 = Not familiar	Option 2	Option 3	Option 4	5 = Very Familiar
Statement 1	☐	☐	☐	☐	☐

3. The device appears suitable for point-of-care (bedside) use. *

☆☆☆☆☆

4. The device appears easy to handle and operate. *

☆☆☆☆☆

5. The design appears ergonomic for prolonged use. *

☆☆☆☆☆

6. The control interface appears intuitive and easy to understand. *

☆☆☆☆☆

7. The device appears capable of adapting to irregular wound geometries. *

☆☆☆☆☆

8. The device appears to have potential to improve current wound treatment methods. *

☆☆☆☆☆

9. What is the most important improvement you would suggest before clinical use?

Enter your answer

Figure 43: Questionnaire screenshot 1

Figure 44: Questionnaire screenshot 2

Ultra-link to the response Excel file: [Click here to view the Project Spreadsheet](#)

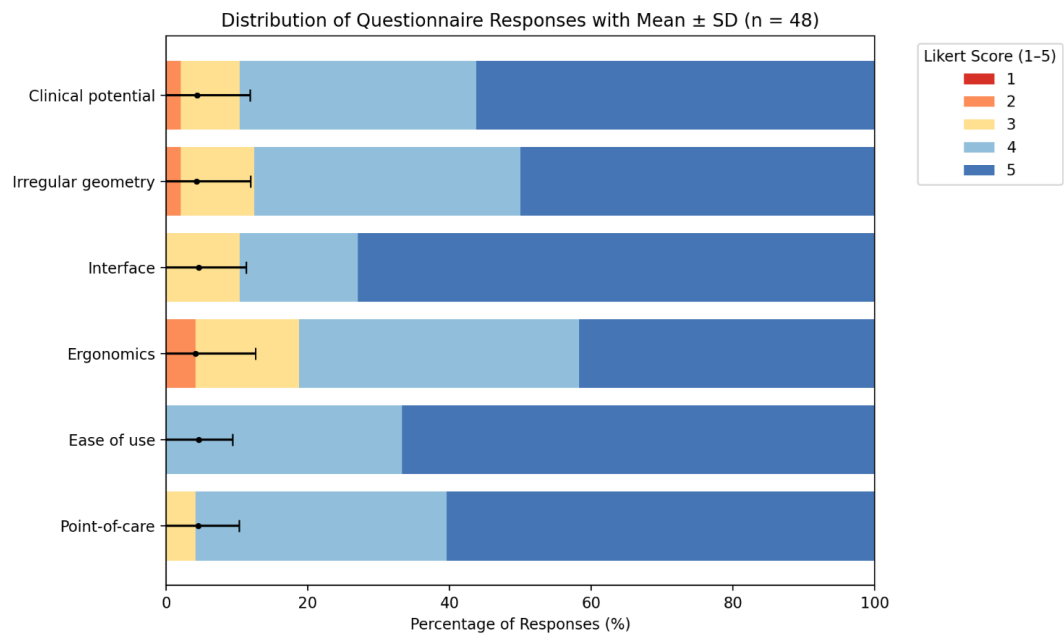


Figure 45: Distribution of questionnaire responses with mean SD (n = 48)

K) Literature Mindmap

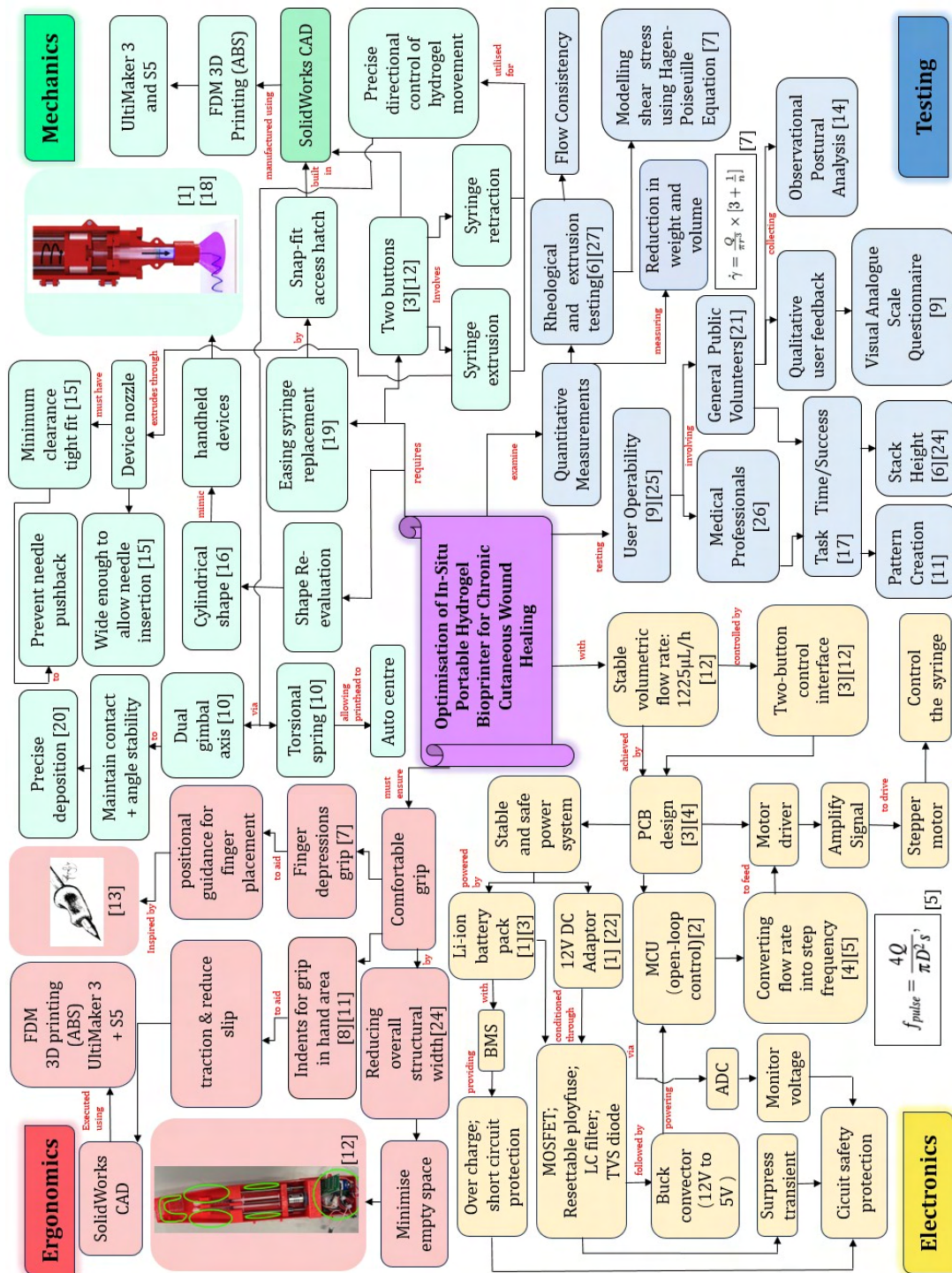


Figure 46: Literature mindmap

L) Risk Assessment

Severity (S)	Occurrence (O)	Detection (D)	Rating
Hazardous without warning	Very high failure is almost inevitable	Absolute uncertainty	10
Hazardous with warning	Very high failure is almost inevitable	Very remote	9
Very high	High repeated failures	Remote	8
High	High repeated failures	Very low	7
Moderate	Moderate occasional failures	Low	6
Low	Moderate occasional failures	Moderate	5
Very low	Moderate occasional failures	Moderately high	4
Minor	Low relatively few failures	High	3
Very minor	Low relatively few failures	Very high	2
None	Remote failure is unlikely	Almost certain	1

Figure 47: Risk assessment criteria [32]

Risk Factor	Consequence	S1	O1	D1	RPN(1)	Preventative Measure Imposed	S2	O2	D2	RPN(2)
<i>Electrical Risk Analysis</i>										
Motor Power Failure	Loss of extrusion and retraction function	8	4	2	64	A battery with a higher capacity was used to ensure steady power supply	8	2	2	32
PCB Board Overheating	Component failure and potential damage to other electrical and mechanical components	8	5	2	80	Heat was dissipated with a copper layout and sufficient airflow over the PCB	7	2	3	42
Accidental overwrite of PCB code	Loss of control functionality and bio-ink extruded at incorrect flow rate	6	4	6	144	ISP port added for proper code updation	6	2	5	60
<i>Mechanical Risk Analysis</i>										
Accidental activation while device handling	Bio-ink material wastage and contamination	5	4	3	60	The activating button must be long pressed to begin extrusion	5	2	1	10
Nozzle Clogging	Sudden increase in pressure leading to over-extrusion of material	6	5	4	120	Integration of pressure sensors with automatic cut-off systems to stop extrusion if clogs are detected	6	3	2	36
Component Slot Mismatch	Assembly issues leading to integration errors affecting device performance	5	7	1	35	Multiple iterative prototypes produced using PLA to validate component fit	5	3	1	15

Table 8: FMEA Risk Analysis for Components.

M) Clinical Consultation Questions

The following questions were discussed with a consultant doctor to evaluate the potential clinical applicability of the handheld bioprinter:

1. What are the main current challenges in treating complex or non-healing wounds?
2. Would a handheld device capable of depositing biomaterials directly onto a wound be clinically useful and in what settings?

3. How important are portability, ergonomics, and intuitive operation for a device used at the point of care?
4. What concerns would you have regarding sterility, infection control, and safe clinical use?
5. Would the ability to adapt to irregular wound geometries provide clinical value?
6. What barriers might limit adoption and what features would be essential for routine clinical use?

N) Bill of Materials

Ultra-link to the Excel file: [Click here to view the Project Spreadsheet](#)

O) Project Management

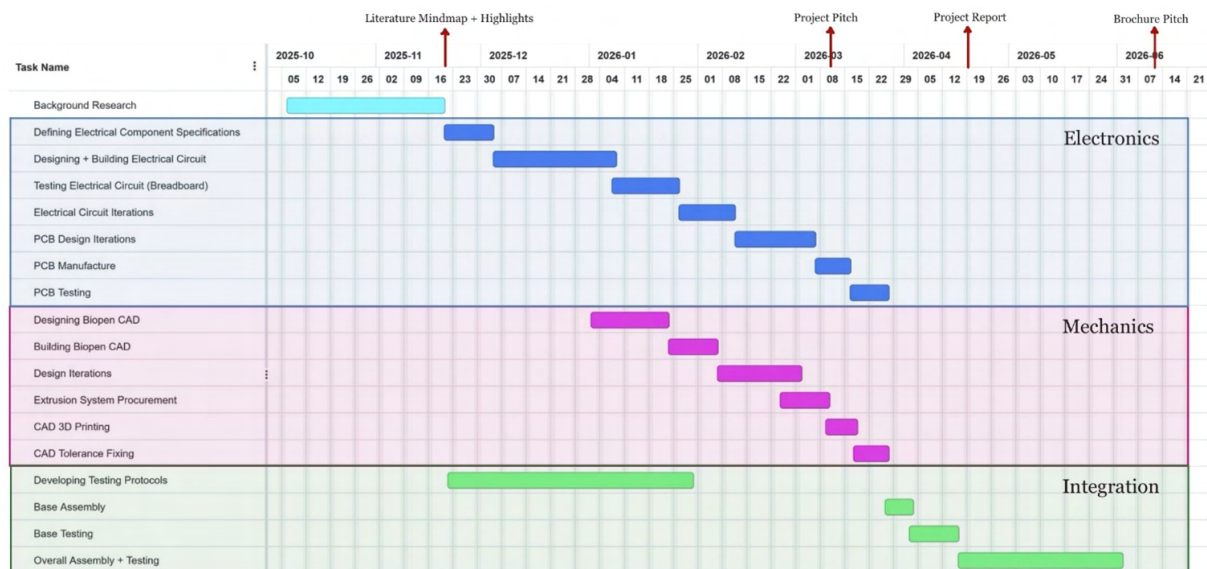


Figure 48: Gantt chart

Figure 49: Tasks completed for each group member

Overall, the project showed consistent progress followed by consistent tracking through weekly internal meetings, alongside initially tri-weekly supervisor meetings that transitioned into bi-weekly meetings once the project entered the design and prototyping stage. All milestones and task allocations were managed via Trello and Microsoft Teams, while group members consistently used WhatsApp for daily updates. Figure xx illustrates the contributions for each milestone of the project; all academic submissions were a group contribution, with Ella and Kennya worked extensively on the Project Pitch.

The project's progression largely followed the original Gantt Chart, although several technical and logistical hurdles required us to adjust the final testing phase. A departure from the original plan was the fact that we obtained unexpected fabrication errors during the PCB manufacturing process, requiring us to reorder and extending the lead time for delivery and individual testing. Moreover, we spent more time prototyping components that required a tight fit due to limitations in the 3D printer's printing uncertainty. To mitigate the impact of these setbacks on the overall project delivery, we prioritised the validation of the core system, and we verified the base's full electronic functionality, whereas the overall assembly has been pushed to the summer term. Despite these hurdles, the overall team progress was not significantly impacted, as our team remained flexible and enthusiastic.

The key project management lessons learned are:

1. Collaboration Across Different Academic Disciplines: Our team was comprised of not only MEng Biomedical Engineers, but also iBSc intercalated Medicine students. This diversity allowed us to approach the project from different perspectives; for example, we were able to get more detailed

clinical feedback and perspective of our proposed product. However, this also introduced challenges in aligning expectations and working styles. As the project is very interdisciplinary, it required us to divide the workload into multiple sub-groups and stages to maximise efficiency. This also meant that after working on a sub-group for a long time, those specific members become difficult to replace when they are unavailable to work, and other members sometimes did not fully understand the parts of the project they were not directly responsible for, occasionally leading to confusion and miscommunication. Therefore, group meetings should focus on bringing everyone up to speed on the overall progress, rather than only presenting individual updates.

2. Flexibility and Buffer Management: While our initial Gantt Chart was comprehensive, it did not account for “buffer zones” to mitigate unexpected delays, such as the PCB fabrication errors. It also did not take into account lab closure days for days where testing is supposed to take place, due to the lack of information regarding Hackspace closure dates until closer to the holidays. However, we came to understand that delays and unpredictability are part of the natural progress of any project, and that managing these issues flexibly and keeping a clear head is the only way to overcome this once you reach this point. We focused on progressing with tasks that were not affected by the delays, such as CAD refinement and rapid prototyping, and although the initial delay was not welcome, it resulted in better iterations of CAD components. In future projects, it would be extremely beneficial to have built-in buffer zones in our timeline so that a delay will not impact the project’s progress at all.

3. Clear Communication: Thorough the project, effective communication was essential to keep the project moving forward; whenever issues or delays occurred, like the PCB fabrication errors or lab closures, it was communicated with all the members in the group quickly so that we could collectively agree on the change of plan without internal confusion. We found that a tiered communication strategy was the most beneficial; Trello and Microsoft Teams allowed everyone to access necessary documentation, WhatsApp provided a platform for quick updates or requests, the weekly internal meetings updated the rest of the team on everyone else’s progress, and the supervisor meetings allowed us to present our progress as a whole, which at the same time also consolidated our own knowledge.